

From Reservoir Replay to Critic-Free Imagined Rehearsal in Sequential RLScape

A chronological experimental chapter on retention, model semantics,
critic failure, and controlled actor recovery

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Continual Reinforcement Learning Research Notes

Experiments completed August 2026; expanded report generated 15 August 2026

Abstract

This chapter records five linked studies in a goal-conditioned DreamerV3 agent operating in RLScape. Study I compares a 200k-capacity global FIFO replay against a persistent, goal-stratified episode reservoir while one agent learns *kill goblin*, *bury bones*, and *chop logs* for 500k environment steps each. Reservoir replay produced strong but transient balanced retention: at 1.25M steps its worst sampled-policy goal reached 49% versus 8% for FIFO, but by 1.5M all reservoir goals had collapsed to at most 16%. Study II finds accurate factual completion recognition, weak one-hot goal semantics, a severely optimistic critic, and a six-step IR target that is 94.4%–97.3% critic bootstrap. Study III causally isolates the actor: critic-free reward-only IR recovers a corrupted policy, reaching 73.0% sampled success at $H = 15$ and 85.3% deterministic success at $H = 20$. Study IV transfers this mechanism to natural continual-learning damage. At the collapsed 1.5M checkpoint, reward-only $H = 15$ recovers bury bones from 14% to 91% sampled and from 4% to 86% deterministic, while standard IR remains unsafe. Study V locates the critic failure. On the same 48,299 real posterior states, mean value bias grows from +1.184 at 1.25M to +1.595 at 1.5M; the slow critic copies this error, imagined targets plateau after $H = 3$, and actor-selected rollouts further prefer high-bootstrap, often lower-reward trajectories. Thus actor/model exploitation amplifies rather than originates an already corrupted value landscape. The combined evidence establishes that imagined actor adaptation is viable on synthetic and natural policy deficits, but the current Dreamer critic is not a safe teacher. Bounded value learning, explicit counterfactual goal supervision, and prespecified adaptation gates are the next requirements for a continual-learning algorithm rather than an episode-local recovery mechanism.

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1 Research programme and chronological evidence chain

The long-term research direction is continual reinforcement learning over an RLscape task landscape: train on goals $A \rightarrow B \rightarrow C \rightarrow \dots$, repeatedly evaluate all previous goals, and measure both forgetting and transfer speed relative to a non-IR baseline. The immediate question was narrower. The preceding FIFO experiment suggested that later phases remove most early-task data, potentially weakening the goal-conditioned reward model and value function that IR relies on. We therefore asked:

1. Does goal-stratified reservoir replay preserve earlier goal competence better than global FIFO replay?
2. Does a retained multi-goal world model and critic make actor-only IR useful at evaluation time?
3. Can the system train continuously for 1.5M steps with research-grade checkpoint and evaluation integrity?
4. If critic-free rehearsal repairs synthetic actor corruption, does it also recover behavior lost naturally during sequential training?
5. Does critic error begin on real posterior states, emerge with imagination depth, or arise because the actor exploits the learned model?

The first experiment contributes a fully persisted three-goal sequential run, six immutable training checkpoints, 24 paired frozen/IR evaluation units, two action-selection modes, and a direct comparison to the previous FIFO run. It is an exploratory, single-seed systems experiment rather than a population-level statistical claim. The later experiments deliberately reuse the same pinned checkpoints. This converts a costly multi-day training run into a sequence of increasingly controlled mechanism tests.

The chapter is organized as a research diary without sacrificing paper-style separation between method, result, and interpretation:

1. **Study I — continual training and replay.** Establish whether reservoir replay preserves usable cross-goal knowledge and whether ordinary actor-only IR helps at checkpoints.
2. **Study II — subsystem and imagination audits.** Hold the replay-state bank fixed and locate whether failure lies in factual event recognition, counterfactual goal semantics, continuation, critic calibration, or the exact IR objective.
3. **Study III — controlled actor corruption and recovery.** Hold the world model and critics fixed, damage only the actor, and causally ablate horizon, entropy, and critic bootstrap.
4. **Study IV — natural forgetting recovery.** Apply the successful critic-free objective to unmodified, naturally forgotten checkpoint actors and measure recovery and harm against paired frozen baselines.
5. **Study V — critic provenance and calibration.** Evaluate identical real posterior states, recorded-action imagination, and actor-selected imagination to identify where critic optimism originates and how it changes with horizon.

To keep implementation artifacts and thesis terminology aligned, the canonical repository stages map to the paper studies as follows. Historical run names beginning with `m` denote engineering milestones and are retained only for provenance.

Table 1: Canonical experiment identifiers used by the repository registry.

Canonical ID	Experiment	Chapter location
Stage 0	Sequential reservoir baseline and checkpoint IR	Study I
Stage 1A	Same-state goal-conditioned head audit	Study IIa
Stage 1B	Exact imagination-path audit	Study IIb
Stage 2	Controlled actor corruption and recovery	Study III
Stage 3A	Critic provenance and calibration audit	Study V
Stage 3B	Natural forgetting recovery	Study IV

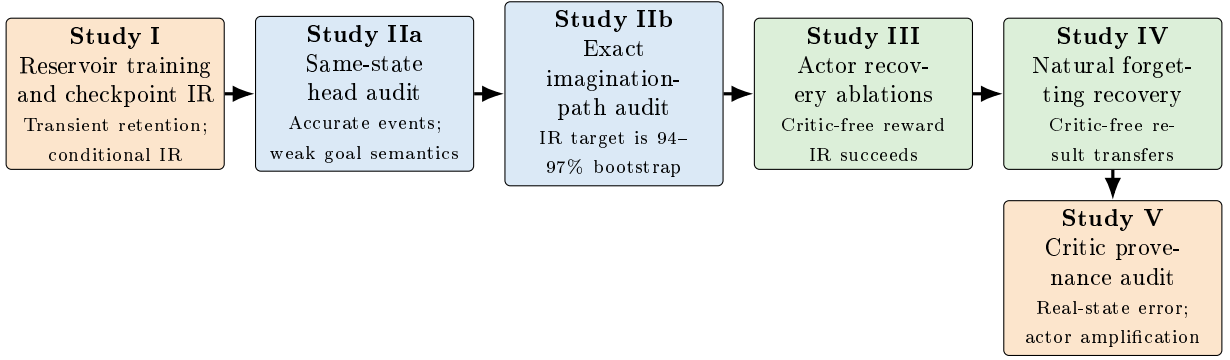


Figure 1: Chronological experimental logic. Each study was designed after observing the previous result; Studies II–V reuse immutable checkpoints and therefore isolate mechanisms without another continual-training run.

2 Study I: sequential reservoir replay and checkpoint IR

2.1 Task sequence

The agent trained on three goals in a fixed curriculum:

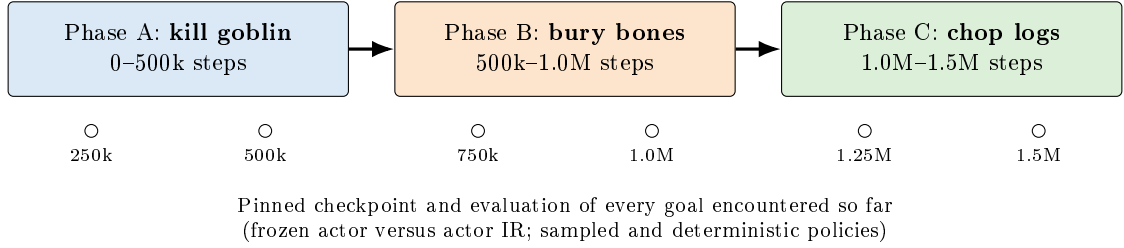


Figure 2: Sequential curriculum and evaluation ladder. There is no return phase in this experiment. The checkpoints permit later ablations without retraining the full curriculum.

Each episode was capped at 200 actions and used sparse binary task-completion reward. The action interface was a flat 28×18 click grid, giving 504 discrete click locations. This was intentionally held fixed between replay conditions so the comparison isolates replay, not environment design.

2.2 Agent and training configuration

The implementation uses DreamerV3 [3] with the repository’s “50m” preset. The observed trainable parameter count was approximately 77.3M; the preset name is therefore a configuration label, not the measured parameter total. The recurrent state-space model (RSSM) uses a deterministic state of width 4,096 and a 32×32 categorical stochastic state. Training used batches of 8 sequences, sequence length 32, imagination horizon 15, training ratio 32, and bfloat16 computation on a 24,GB GPU.

Let o_t be the image, a_t the click action, and $s_t = (h_t, z_t)$ the Dreamer latent state. The goal-agnostic dynamics are

$$h_t = f_\phi(h_{t-1}, z_{t-1}, a_{t-1}), \quad (1)$$

$$z_t \sim q_\phi(z_t \mid h_t, e_\phi(o_t)) \quad \text{during representation learning}, \quad (2)$$

$$z_t \sim p_\phi(z_t \mid h_t) \quad \text{during imagination}. \quad (3)$$

Five RLScope goals are represented by a five-dimensional one-hot vector g_t , although only three goal IDs occur in this experiment. The behavior feature is

$$x_t = [h_t; z_t; g_t]. \quad (4)$$

The reward, continuation, actor, value, and slow-value heads consume x_t :

$$\hat{r}_t \sim p_\phi(r_t \mid h_t, z_t, g_t), \quad \hat{c}_t \sim p_\phi(c_t \mid h_t, z_t, g_t), \quad (5)$$

$$a_t \sim \pi_\theta(a_t \mid h_t, z_t, g_t), \quad v_\psi = v_\psi(h_t, z_t, g_t). \quad (6)$$

The RSSM transition and visual representation remain goal-agnostic. Thus the shared latent should model what is happening on screen, while the heads interpret that state under the requested goal.

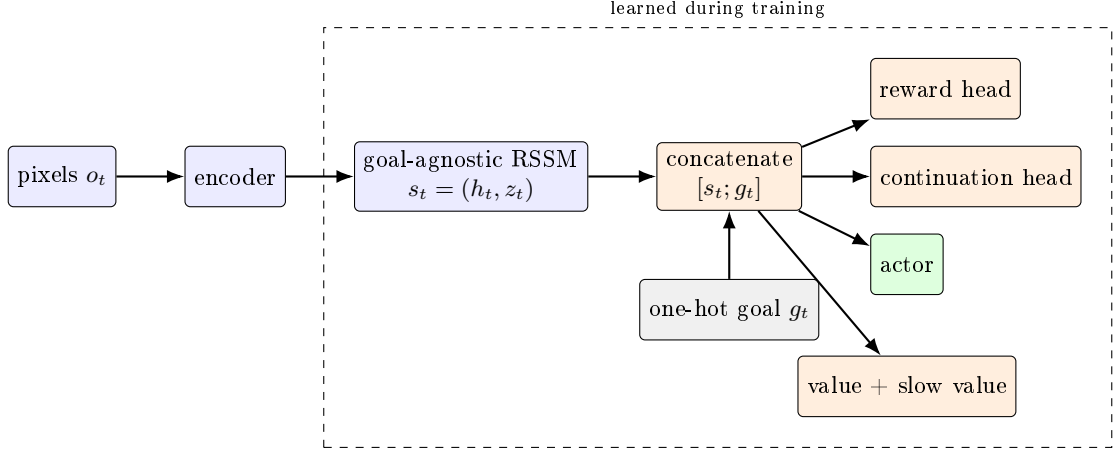


Figure 3: Goal-conditioning architecture. During evaluation-time IR, only the actor parameters (green) are updated; the world model, reward/continuation heads, and critic/value functions are frozen.

2.3 Replay conditions

FIFO baseline. The comparison run used the original global replay with capacity 200,000 sequence starts. Once full, newly collected experience evicts the oldest experience regardless of goal. Uniform training batches therefore increasingly reflect the current phase.

Goal-stratified episode reservoir. The new replay stores at most $M_g = 1,000$ complete episodes independently for each represented goal, for a total capacity of 3,000 episodes. Within goal g , the n_g th observed episode is processed by Algorithm R [4]:

$$\Pr(\text{admit episode } n_g) = \min \left(1, \frac{M_g}{n_g} \right). \quad (7)$$

If admitted after the group is full, it uniformly replaces one of the M_g resident episodes. Consequently every historical episode for a goal has equal probability M_g/n_g of residing in that goal’s reservoir. Training samples a represented goal uniformly, an episode uniformly within that goal, and then a random sequence start. Short fragments are concatenated with explicit `is_first` boundaries so the recurrent state resets correctly.

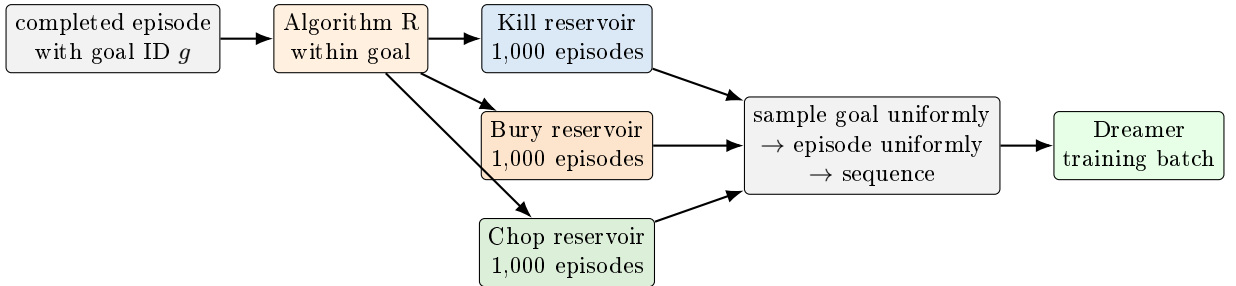


Figure 4: Goal-stratified episode reservoir. Statistical retention is uniform over the episode stream within each goal; optimization sampling is balanced across represented goals.

This balance changes the effective emphasis on new data. The global training ratio remains 32, but the expected share of updates drawn from the current goal is 1, 1/2, and 1/3 in phases A, B, and C. In this limited sense, current-goal exposure is approximately 32, 16, and 10.7 replayed samples per environment step. This is a plausible source of slower acquisition or instability and is important when interpreting the comparison.

2.4 Evaluation-time imagined rehearsal

IR performs temporary test-time optimization of the actor from the agent’s current latent state. For each environment action, Monte Carlo Posterior Batching (MCPB) draws $N = 128$ posterior latent samples, imagines $H = 6$ steps under the frozen world model, and takes one actor gradient update with learning rate 4×10^{-5} . MCPB is posterior batching: it does not denote particles and does not perform particle filtering, weighting, or resampling. The objective is the Dreamer actor objective: increase imagined, critic-estimated return while retaining the configured entropy bonus $\eta = 3 \times 10^{-4}$,

$$\theta' \leftarrow \theta - \alpha \nabla_{\theta} \left[-\mathbb{E}_{\hat{\tau} \sim p_{\phi}, \pi_{\theta}} \left(\hat{A}_{\psi}(\hat{\tau}) + \eta \mathcal{H}[\pi_{\theta}(\cdot \mid \hat{s}, g)] \right) \right]. \quad (8)$$

The precise implementation uses the repository’s normalized imagined-advantage actor loss. The world model, reward/continuation models, critic, and slow critic are frozen. Actor updates accumulate within an episode, are discarded at the episode boundary, and never alter the saved training checkpoint. There is no confidence, entropy, or value-based gate: IR runs at every action from the first action ($k = 0$).

Conceptually, IR asks whether the retained model and critic contain enough goal-specific knowledge to repair a locally damaged actor without collecting gradient updates from the real reward stream. It is not hindsight relabeling [1]: replay goals and rewards are not relabeled.

2.5 Evaluation protocol and integrity

At 250k-step intervals, every goal encountered by that point was evaluated. For each checkpoint–goal cell we ran:

- 100 frozen and 100 IR episodes with sampled actions;
- 50 frozen and 50 IR episodes with deterministic actions.

Frozen and IR conditions used matching seed schedules and goal requests. Across 12 checkpoint–goal cells this yielded 3,600 episodes in 48 complete conditions (24 frozen/IR pairs). Requested episode counts, goal IDs, reset counts, return codes, and checkpoint parameter digests passed the evaluator’s integrity checks. Because frozen and IR conditions ran in separate RLScope process lifetimes, their initial image hashes were not identical despite matching seed/goal identities. The analysis therefore treats them as seed-paired aggregate comparisons, not guaranteed pixel-identical counterfactual trajectories.

The FIFO evaluation predates a fix to the strict frame-hash pairing checker: its queue recorded failure even though individual condition summaries completed. We use the complete first attempts at checkpoints through 1.5M steps descriptively. This caveat does not affect the reservoir run’s own completion status.

3 Study I results

3.1 Execution and systems outcome

The reservoir experiment completed all six training milestones and all 24 paired evaluation units without a training restart. Training took approximately 62.3 hours, evaluation 19.7 hours, and the full workflow approximately 82 hours. This satisfies the operational objective: a multi-day continual experiment can train, checkpoint, resume evaluation, and terminate cleanly.

At 1.5M steps the reservoir contained exactly 1,000 episodes per goal, 18,574 historical episodes had been considered for admission, and the retained episodes comprised 265,802 transitions. The latest sampling interval drew 655, 652, and 693 batches from kill, bury, and chop respectively, demonstrating near-uniform goal selection. Replay occupied 28.5, GiB of host memory and the full process 34.3, GiB, leaving 22.9, GiB available on the 62, GiB machine. Acting and training throughput were approximately 6.66 and 213 steps/s. The FIFO endpoint used 23.2, GiB for replay and 26.5, GiB for the process, with 6.92 acting steps/s and 222 training steps/s. The methods are therefore not memory-matched.

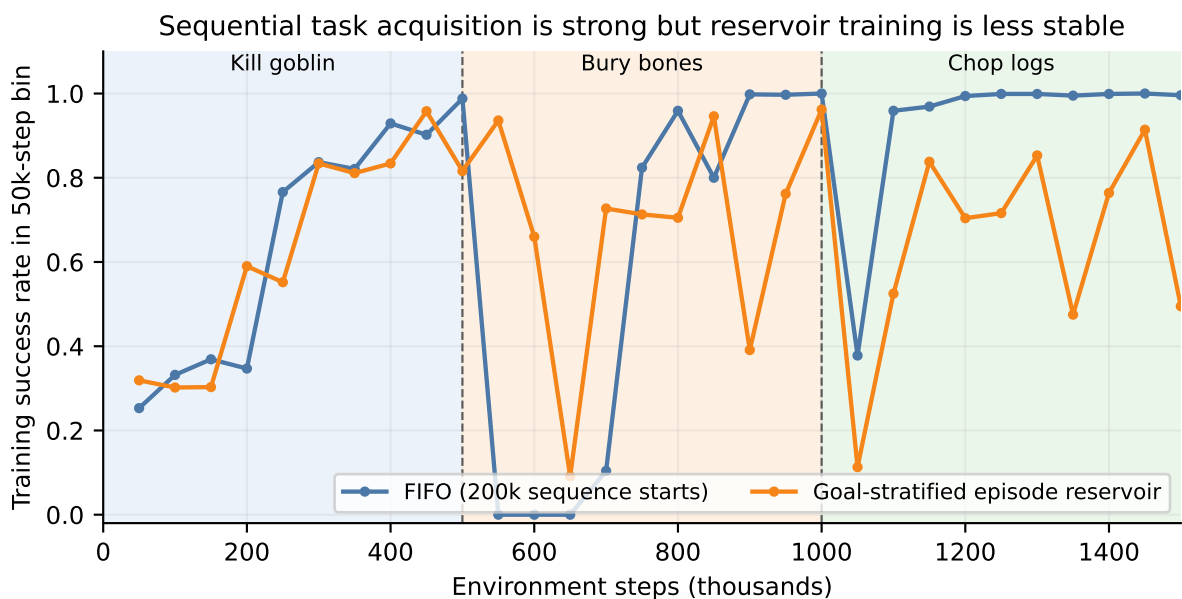


Figure 5: Training success in successive 50k-step bins. FIFO learns each active task cleanly after an initial delay. Reservoir replay reaches high success on every phase but exhibits repeated large collapses, including late in the final phase. These are behavioral training metrics, not held-out evaluation.

Reservoir training reached rolling 100-episode peaks of 99%, 100%, and 100% on kill, bury, and chop, proving that the architecture and action interface can solve every chosen goal. Final rolling success was only 85%, 54%, and 16%. In the last 10k steps of each phase, success was 84.2%, 37.5%, and 11.3%. The main failure is therefore retention/stability, not basic reachability.

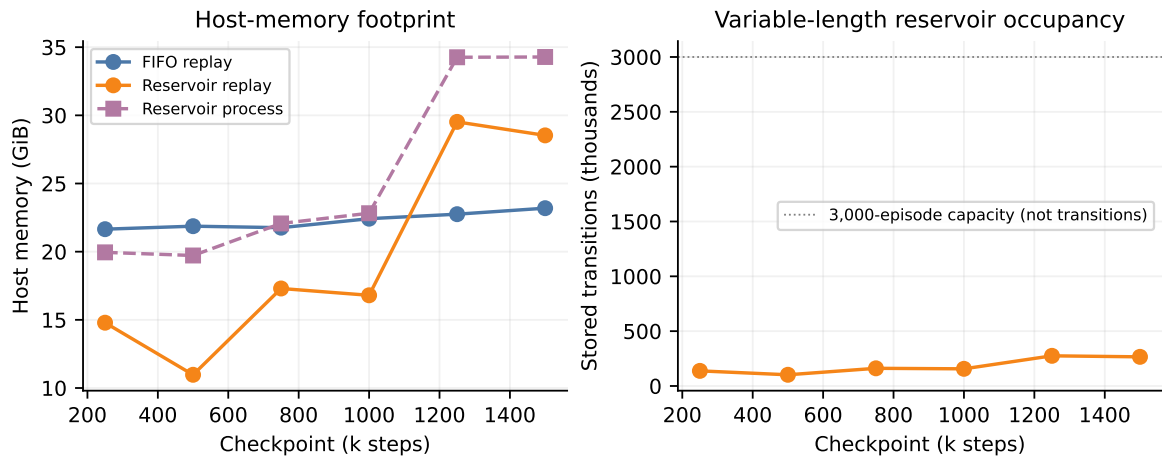


Figure 6: Resource use at pinned milestones. Episode capacity does not fix transition count because episode lengths vary. The 3,000-episode reservoir ended larger than the 200k-start FIFO buffer, making the comparison intentionally algorithmic rather than memory-matched.

3.2 Frozen-policy checkpoint matrix

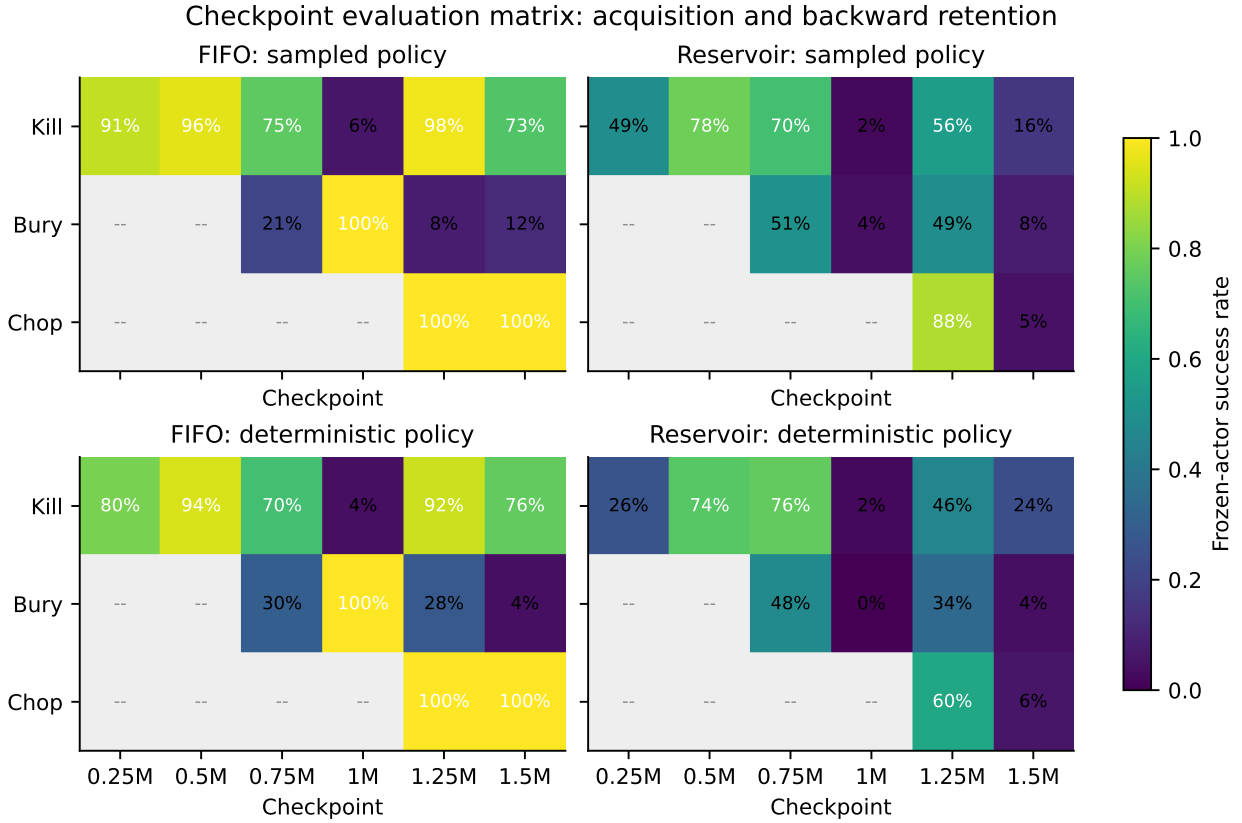


Figure 7: Frozen-actor success at all evaluable checkpoint–goal cells. Gray cells correspond to goals not yet introduced. The reservoir temporarily improves balanced retention but undergoes synchronized collapses at 1.0M and 1.5M steps.

Across all available cells, FIFO frozen success averaged 65.0% sampled and 64.8% deterministic. Reservoir frozen success averaged 39.7% and 33.3%, a substantial overall deficit. On acquisition checkpoints—the end of the phase in which a goal was introduced—reservoir trailed FIFO by 38.8 points sampled and 48.3 points deterministic. Goal-balanced replay is therefore not a free retention improvement: it materially reduces or destabilizes current-task learning under the present update schedule.

The result is not uniformly negative. At 750k steps, after only 250k bury-bones training, reservoir sampled success was 70% on kill and 51% on bury; FIFO achieved 75% and 21%. The reservoir’s worst task was therefore 30 points better. At 1.25M, after 250k chop training, reservoir simultaneously achieved 56%, 49%, and 88%; FIFO achieved 98%, 8%, and 100%. The reservoir’s worst task was 49% rather than 8%. This is the clearest positive evidence: stored old-goal episodes, goal-balanced sampling, and goal-conditioned heads can jointly recover multiple skills while a new goal is being learned.

That balance was transient. At 1.0M, reservoir kill and bury collapsed to 2% and 4%; at 1.5M all three goals were 16% or worse. FIFO showed selective forgetting instead: its final sampled scores were 73%, 12%, and 100%. FIFO’s retained mean was much higher, but its weakest old goal remained poor.

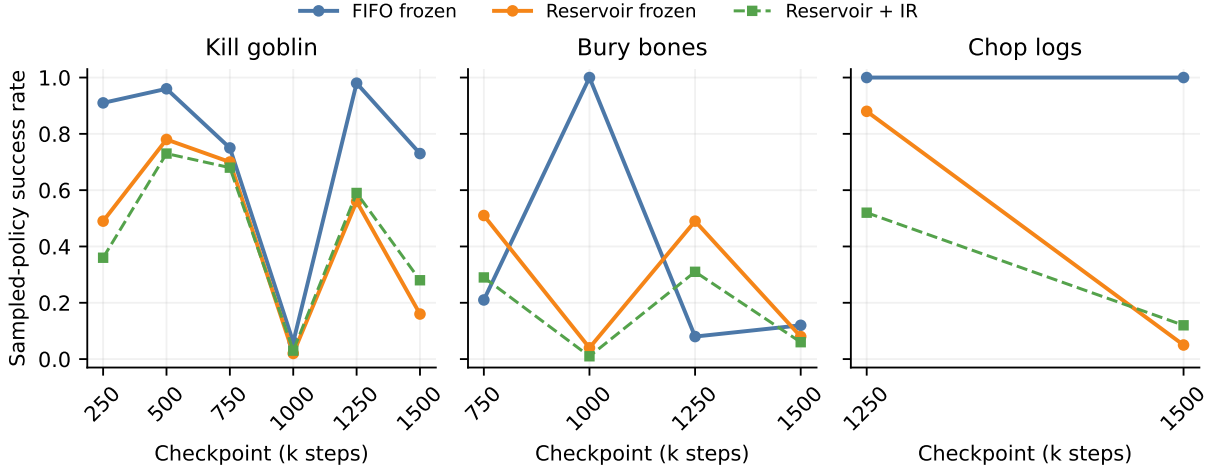


Figure 8: Sampled-policy trajectories for each goal. The dashed green line adds actor IR to the reservoir checkpoint. Reservoir skill recovery between 1.0M and 1.25M is followed by broad collapse between 1.25M and 1.5M.

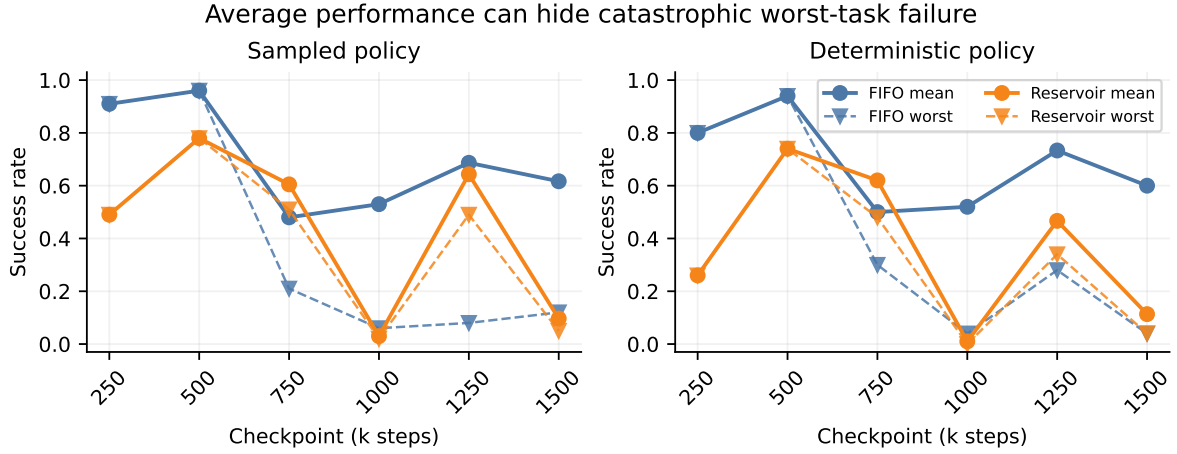


Figure 9: Mean and worst encountered-goal performance. Reservoir replay has useful mid-phase worst-task advantages, while both methods can hide severe forgetting behind a reasonable mean.

3.3 Forgetting

Forgetting is reported as the maximum checkpoint success for a goal minus its final-checkpoint success. In the sampled reservoir evaluations it was 62 points for kill, 43 for bury, and 83 for chop. FIFO forgetting was 25, 88, and 0 points. Deterministic reservoir forgetting was 52, 44, and 54 points, versus 18, 96, and 0 for FIFO. Reservoir failure was broad; FIFO failure was concentrated in bury bones.

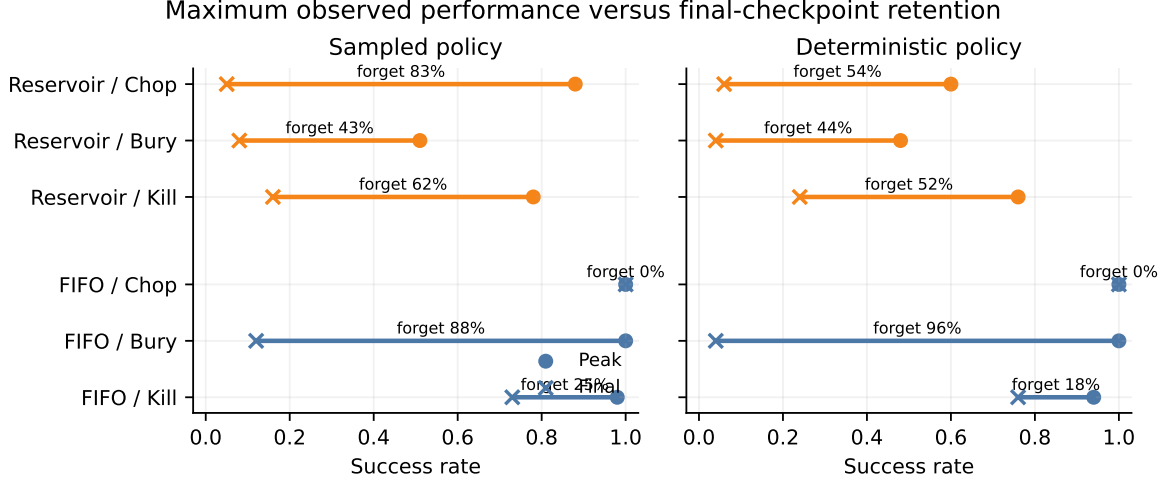


Figure 10: Peak-to-final forgetting. Circles mark the best observed checkpoint and crosses mark the final checkpoint. A useful continual learner should keep both markers high and close together.

3.4 Effect of iterative rehearsal

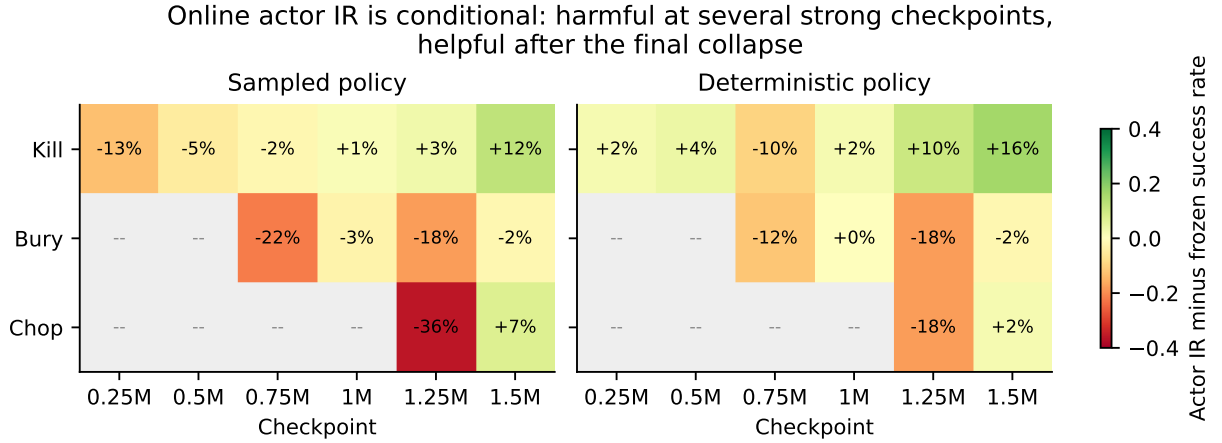


Figure 11: IR effect in the reservoir run. Positive values mean actor-only IR improved success over the frozen checkpoint. Strong harm is concentrated at several high-performing checkpoint-goal pairs; final-checkpoint effects are small but consistently positive for kill and chop.

Across 24 mode-specific comparisons, IR improved 10, tied 1, and harmed 13. Averaged over all reservoir checkpoint-goal cells, sampled success fell from 39.7% to 33.2% (−6.5 points), while deterministic success fell from 33.3% to 31.3% (−2.0 points). The strongest harms were sampled chop at 1.25M (−36 points), sampled bury at 750k (−22), and both sampled and deterministic bury at 1.25M (−18).

At the final checkpoint—the most directly relevant corrupted-policy case—the pattern reversed. Mean sampled success rose from 9.7% to 15.3%, and deterministic success from 11.3% to 16.7%. Sampled kill improved from 16% to 28% (paired bootstrap 95% interval for the 12-point change: [2, 23] points); chop improved from 5% to 12%; bury fell from 8% to 6%. Deterministic kill improved from 24% to 40%, although its 50-episode interval was wide ([−4, 36] points).

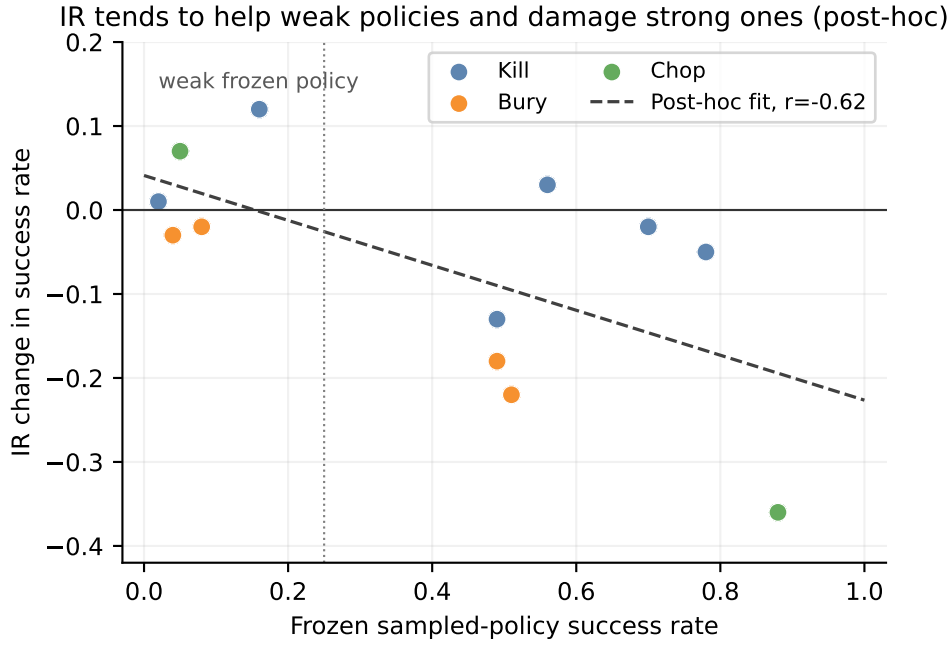


Figure 12: Post-hoc relationship between frozen policy strength and sampled-policy IR effect. The correlation is $r = -0.50$. For cells with frozen success at most 25% ($n = 10$), mean IR effect was +3.3 points and six improved; above 25% ($n = 14$), mean effect was -9.6 points. This threshold was selected after seeing the data and is a hypothesis for a future gate, not a confirmatory result.

The negative association supports the original intuition behind gated IR: adapt when the actor appears damaged, and avoid perturbing a competent policy. It does not validate entropy as the gate variable, because this experiment did not record or intervene on a prespecified entropy threshold. Entropy also remained in the IR objective and may have dominated short-horizon updates in some states.

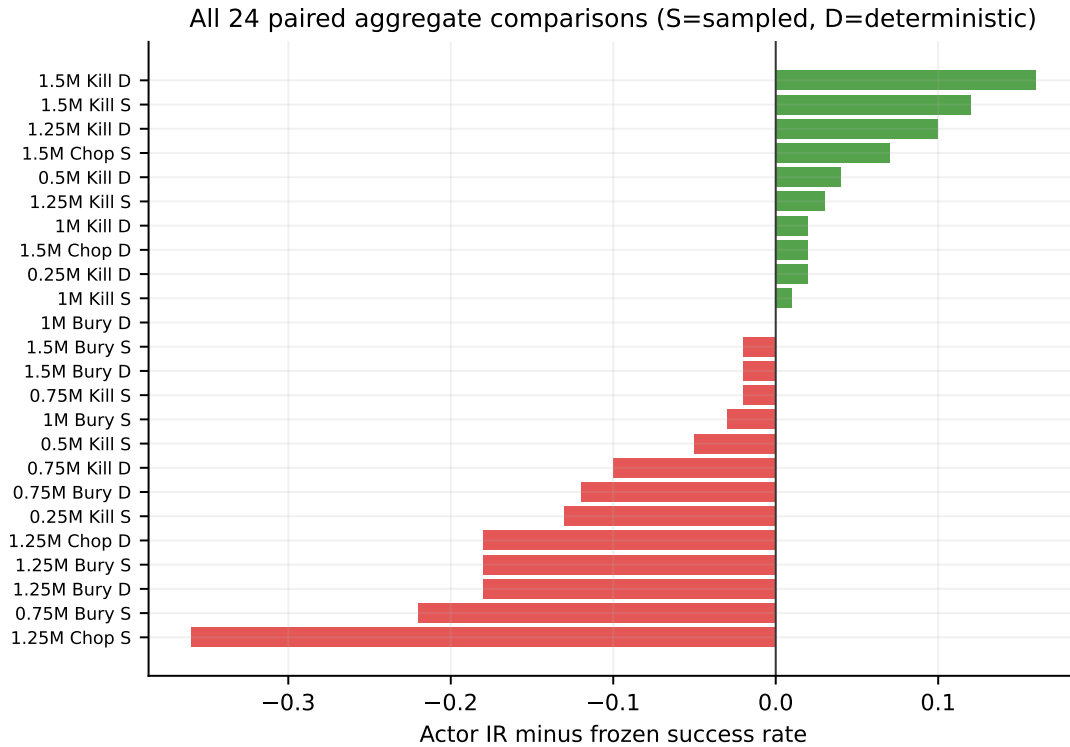


Figure 13: Every reservoir IR comparison, sorted by effect. S and D denote sampled and deterministic evaluation. The distribution is visibly asymmetric: a few large harms dominate several modest gains.

Table 2: Aggregate success rates. Values are unweighted means over available checkpoint–goal cells.

Replay	Policy mode	Frozen	Actor IR	IR change
FIFO	Sampled	0.650	0.620	−0.030
FIFO	Deterministic	0.648	0.623	−0.025
Reservoir	Sampled	0.397	0.332	−0.065
Reservoir	Deterministic	0.333	0.313	−0.020
Reservoir, final only	Sampled	0.097	0.153	+0.056
Reservoir, final only	Deterministic	0.113	0.167	+0.053

4 Study I interpretation

4.1 What reservoir replay did establish

The experiment rules out the claim that old-task data are wholly inaccessible under the new workflow. All goal strata were full, goal sampling was balanced, and earlier policies recovered sharply during chop training: from 1.0M to 1.25M, sampled kill rose from 2% to 56%, bury from 4% to 49%, while chop reached 88%. That combination would be very unlikely if replay groups, reset boundaries, goal IDs, or head conditioning were completely broken.

The experiment does *not* prove that the reward model and critic are well calibrated for every goal. Behavioral success is an end-to-end diagnostic. A model can contain enough signal for recovery while still having biased reward predictions, value misranking, or compounding imagination error. Direct head-calibration probes remain necessary.

4.2 Why balanced replay may still collapse

Several mechanisms are consistent with the observed synchronized swings:

1. **Optimization interference.** Shared encoder, RSSM, and large actor receive gradients induced by multiple visually similar but behaviorally distinct goals. Balanced data availability does not guarantee compatible gradients.
2. **Reduced current-task dose.** Once three strata exist, only about one third of replay batches target the active goal, while the nominal training ratio remains fixed. The reservoir condition is not update-dose matched to FIFO.
3. **Episode-uniform versus transition-uniform sampling.** Complete-episode selection changes the weight of short successful and long failed episodes relative to FIFO sequence-start sampling.
4. **Head/world-model drift.** The RSSM is goal-agnostic and shared. Even if every goal is represented in replay, representation changes can move the inputs seen by reward, value, and actor heads.
5. **Single-run stochasticity.** With one seed, a late policy collapse cannot yet be separated from an unlucky optimization trajectory.

The result therefore argues for measuring subsystem health, not merely increasing replay capacity.

4.3 Why ungated IR can hurt

IR trusts imagined returns. When the actor is already competent, one update at every real action creates many opportunities to move away from a successful action mode because of value error, model exploitation, posterior sampling noise, or the entropy term. A six-step horizon reduces compounding error but does not eliminate it. Conversely, a badly damaged actor has more room for improvement, which matches the final-checkpoint recovery and the post-hoc negative correlation in Figure 12.

The next IR study should therefore begin with a controlled actor-corruption experiment: choose a checkpoint with high task success and intact goal-conditioned heads, perturb only the actor to known severity levels, and measure recovery under frozen versus adapted actors. That design isolates whether IR can repair policy damage when its model and critic are known to be competent.

5 Study I limitations and threats to validity

- **One seed.** All uncertainty intervals quantify episode-level evaluation variability, not training-seed variability. They must not be interpreted as generalization across runs.
- **Replay conditions are not capacity matched.** FIFO capacity is counted in sequence starts; reservoir capacity is counted in complete episodes and consumed more host memory.
- **Optimization exposure differs.** Goal-balanced sampling necessarily decreases the current task’s fraction as goals accumulate. A fair algorithmic ablation may need either ratio compensation or a tunable current/old mixture.
- **Evaluation pairing is imperfect.** Seed and goal schedules match, but separate process lifetimes produced different initial frame hashes. Episode-order paired bootstrap intervals [2] are informative, not exact paired-environment causal estimates.
- **FIFO summaries are legacy artifacts.** The old evaluation supervisor marked the queue failed under a subsequently fixed frame-hash rule. Completed first-attempt summaries are retained for descriptive comparison.
- **No direct model calibration.** Reward-head accuracy, value calibration, imagined-return bias, and goal separability were not measured. Behavioral success alone cannot locate the failing subsystem.
- **Post-hoc gate analysis.** The 25% frozen-performance split and its correlation were discovered after observing outcomes. A future run must prespecify and validate any gate.
- **Environment scope.** The three chosen tasks use the same 504-way click grid and are solvable with left clicks. Results need not transfer to the two harder RLScape goals or to continuous robotic control.

6 Decision point after Study I

The checkpoint archive makes the following sequence possible without immediately repeating the multi-day run:

1. **Head audit at strong and collapsed checkpoints.** For a fixed image/latent set, measure goal-conditioned reward prediction, continuation, value ranking, and imagined-versus-real return. Include wrong-goal counterfactuals to quantify whether g changes the heads functionally.
2. **Actor corruption and recovery.** Start from the strongest checkpoint for each goal; apply controlled actor noise or partial reinitialization; compare frozen, ungated IR, IR without entropy, and IR with smaller learning rates/horizons.
3. **Prespecified IR gate.** Gate on a measurable signal available at test time—for example ensemble disagreement, low predicted value relative to a checkpoint baseline, or sustained high entropy. Entropy alone should be treated as one candidate, not an assumption.
4. **Replay-mixture ablation.** Keep Algorithm R retention but compare uniform goal sampling with a current-goal-biased mixture, such as 50% current and 50% uniformly old. Match total update dose where possible.
5. **Stability replication.** Repeat the best replay/IR configuration over multiple seeds, then extend from three to all five RLScape goals.
6. **Return phase.** Finally revisit A after $A \rightarrow B \rightarrow C$ to quantify relearning speed, forward/backward transfer, and whether IR reduces real-environment samples.

Hindsight relabeling remains a possible later extension for sparse rewards, but it is not the first diagnosis: all three tasks already reached near-perfect training windows, so the immediate bottleneck is maintaining and exploiting learned competence.

7 Interim conclusion after Study I

The reservoir implementation worked, the long-running workflow was operationally robust, and every task was individually learnable inside the sequential run. Yet goal-stratified reservoir replay did not solve continual learning by itself. It produced compelling but temporary balanced retention, then ended in a synchronized behavioral collapse substantially worse than FIFO’s final mean. Actor-only IR was also not uniformly beneficial: fixed every-action updates often damaged strong actors, while providing modest recovery at the final weak checkpoint.

The most useful outcome is therefore diagnostic. Old experience and goal-conditioned knowledge can support recovery, but neither balanced replay nor ungated test-time adaptation guarantees stability. The next experiments should exploit the saved checkpoints to separate actor damage from reward/value/world-model failure and to design an IR gate before spending another multi-day training budget.

8 Study II: locating the failure inside the learned IR target

Study I left an ambiguity that could not be resolved from behavioral success alone. A weak actor may fail because the policy parameters are damaged while the world model, reward head, continuation head, and critic remain useful. Alternatively, the actor may be reacting rationally to an inaccurate learned objective. This distinction is decisive for IR: test-time actor optimization is only useful when the frozen components supply a trustworthy teaching signal.

Study II therefore compares the **strong 1.25M checkpoint** with the **collapsed 1.5M checkpoint**. At 1.25M, sampled-policy success was 56%/49%/88% on kill/bury/chop. At 1.5M, every task was at most 16%, with a mean of 9.7%. The analysis has two linked parts. Study IIa evaluates the learned heads on real replay trajectories. Study IIb executes the exact actor-IR imagination path, without updating parameters, to measure where its return and gradient signals originate.

8.1 Study IIa method: immutable same-state head audit

The state bank was drawn from the pinned 1.25M replay archive, not recollected separately under each checkpoint. For every trained goal it contains 64 successful and 64 failed episodes, yielding 384 episodes and 48,299 states. Both checkpoints therefore receive the same image sequences, actions, reset boundaries, rewards, continuations, and factual goal labels. This design removes changes in environment sampling from the strong–weak comparison.

Each sequence is encoded under each checkpoint to obtain its posterior latent trajectory. At every state, the reward, continuation, online value, and slow-value heads are queried under all five one-hot goal IDs: the three trained goals and the unused *light fire* and *catch fish* slots. “Factual” means the goal used when that replay episode was collected. “Counterfactual” means that the latent state and every other input are held fixed while only the one-hot goal is changed.

For reward and continuation, the most important counterfactual query is a completion state for goal i evaluated under another goal $j \neq i$. The desired targets are

$$r(s^{\text{complete}(i)}, g_i) = 1, \quad c(s^{\text{complete}(i)}, g_i) = 0, \quad (9)$$

$$r(s^{\text{complete}(i)}, g_j) = 0, \quad c(s^{\text{complete}(i)}, g_j) = 1 \quad (j \neq i). \quad (10)$$

The second line was never supplied by ordinary on-policy replay: every observed completion appears only with its active goal. The audit measures whether one-hot conditioning nevertheless induced that semantic distinction.

Value calibration compares predicted values with recorded discounted returns along the replay trajectory. This is a diagnostic rather than an exact reconstruction of Dreamer’s critic loss: replay actions are off-policy relative to each audited checkpoint, whereas Dreamer trains lambda returns under imagined policy rollouts. The comparison remains valuable because both checkpoints are evaluated on identical states, empirical returns remain in $[0, 1]$, and the strong-to-weak direction can be compared directly. Parameter digests before and after both audits are identical.

8.2 Study IIa results: event recognition without goal semantics

Factual reward prediction remains excellent. Reward AUROC is 0.99993 at 1.25M and 0.99991 at 1.5M. Mean reward prediction on genuine completion states is 0.961 and 0.957, while ordinary-state false positives remain very small. Thus replay retention preserved a highly accurate detector of visible completion-like events through the behavioral collapse.

The reward head scarcely uses the requested goal. On the same completion states, the factual-minus-counterfactual reward margin is only 0.0033 at 1.25M and 0.0018 at 1.5M. Choosing the goal with the largest reward prediction identifies the correct goal only 39.6% and 38.0% of the time despite five goal slots. The head has learned “a task completed” much more strongly than “the requested task completed.”

Continuation has the same semantic failure. The continuation head accurately predicts stopping on factual completion states and continuing on ordinary states. Yet mean stop probability on wrong-goal completion queries is already 93.9%/94.3% at the strong/weak checkpoint, compared with only 95.6%/95.9% for the matching goal. Reward and continuation are therefore separate heads but share the same missing counterfactual supervision.

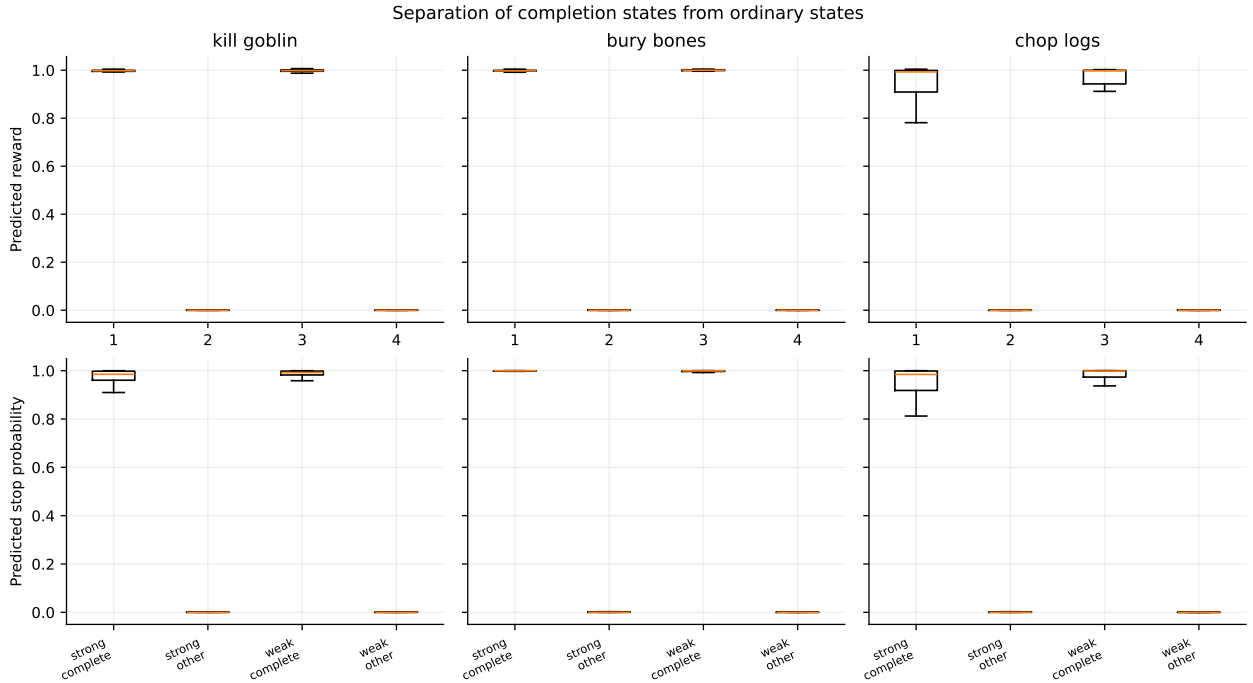


Figure 14: Study IIa factual event recognition. Reward (top) and stopping (bottom) almost perfectly separate completion states from ordinary states at both checkpoints. This figure is deliberately paired with the counterfactual analysis: factual accuracy alone can hide the fact that the same completion state receives almost the same output under every requested goal.

The critic is severely overoptimistic and worsens after collapse. At 1.25M, mean factual critic predictions are 1.62, 1.73, and 0.79 for kill, bury, and chop, whereas empirical discounted-return means are 0.20, 0.10, and 0.21. At 1.5M, values rise further to 1.99, 2.00, and 1.36 even though real policy success collapses. Value-return Pearson correlations deteriorate from 0.182/0.107/0.307 to 0.048/0.064/0.061. At episode starts, value barely separates successful from failed kill trajectories, reverses their ordering for bury, and weakens for chop.

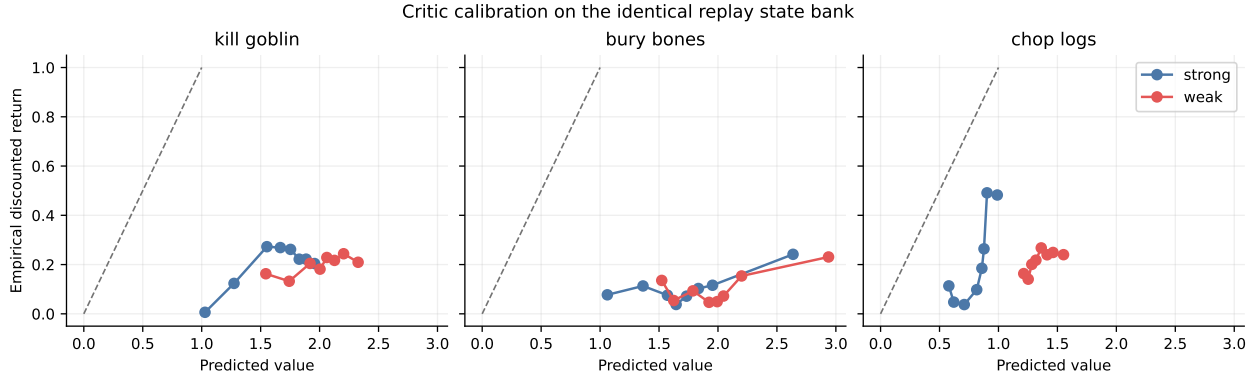


Figure 15: Study IIa critic calibration on the identical replay-state bank. The dashed line is ideal calibration. Predicted values are far above recorded discounted returns, and the weak checkpoint (red) is generally more optimistic despite much worse real behavior.

The slow critic does not provide an independent safety signal. Its mean absolute disagreement with the online critic is only 0.006–0.017; both share the same optimism and drift. Posterior categorical entropy changes only from 0.564 to 0.543 and deterministic-state RMS from 0.256 to 0.262. These scalar summaries do not rule out task-relevant representation drift, but they rule out a gross posterior entropy or magnitude collapse as the immediate explanation.

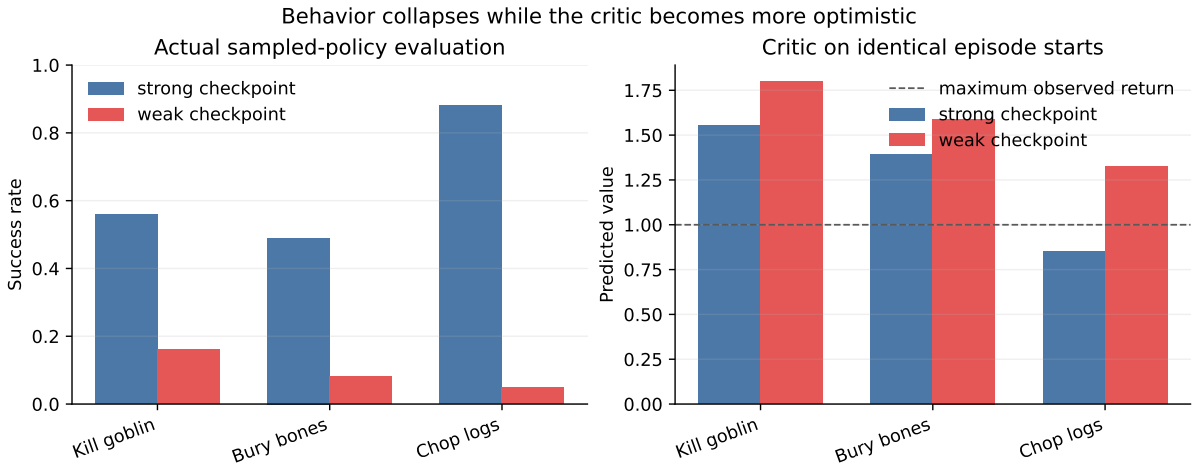


Figure 16: Behavior and critic confidence move in opposite directions between the strong and weak checkpoints. This inverse relationship motivated an audit of the exact imagined return used by actor IR rather than treating critic output as a valid gate.

8.3 Study IIb method: exact six-step imagination-path audit

Study IIb reuses the same 384-episode selection. Four deterministic, nonterminal anchors are requested from each episode; short trajectories yield 1,433 valid anchors. At every anchor, the agent reconstructs the posterior latent state and sweeps all five goal IDs. For each anchor–goal pair, it draws 128 paired posterior samples and executes the same $H = 6$ policy imagination used by Study I actor IR. In total, each checkpoint produces 7,165 anchor–goal rollouts and 5,502,720 imagined action transitions.

No optimizer update is applied. Instead, the audit records, at each imagined time step:

- predicted reward and the incidence of reward-like events;
- continuation and predicted stopping events;
- online and slow critic values;
- lambda return, split into reward-only and critic-bootstrap contributions;

- actor reinforce/advantage term, entropy term, and policy entropy;
- actual replay goal, probed goal, anchor location, and success/failure stratum.

Randomness is paired across goal probes so differences are attributable to the one-hot goal rather than different posterior samples. Parameter digests again match before and after the audit.

For clarity, write the imagined truncated return abstractly as

$$\hat{G}_t^{\lambda,H} = \underbrace{\sum_{k=0}^{H-1} w_{t,k} \hat{r}_{t+k}}_{\text{reward contribution}} + \underbrace{b_{t,H} v_{\bar{\psi}}(\hat{s}_{t+H}, g)}_{\text{critic bootstrap}}, \quad (11)$$

where the actual implementation uses Dreamer’s lambda-return weighting and continuation discounts. The audit reconstructs both contributions rather than inferring them from the total.

8.4 Study IIb results: IR optimizes the critic, not imagined reward

Critic bootstrap dominates the IR target. At the strong checkpoint, mean factual lambda return is 1.450: only 0.072 is attributable to imagined rewards and 1.378 to bootstrap, a 94.4% bootstrap fraction. At the weak checkpoint, mean return rises to 1.868 despite behavioral collapse: reward contribution falls to 0.039 while bootstrap rises to 1.829, or 97.3%. Ordinary anchors are more than 99% bootstrap driven. Even immediately before recorded completion, bootstrap still contributes 69.1% at the strong checkpoint and 82.4% at the weak checkpoint.

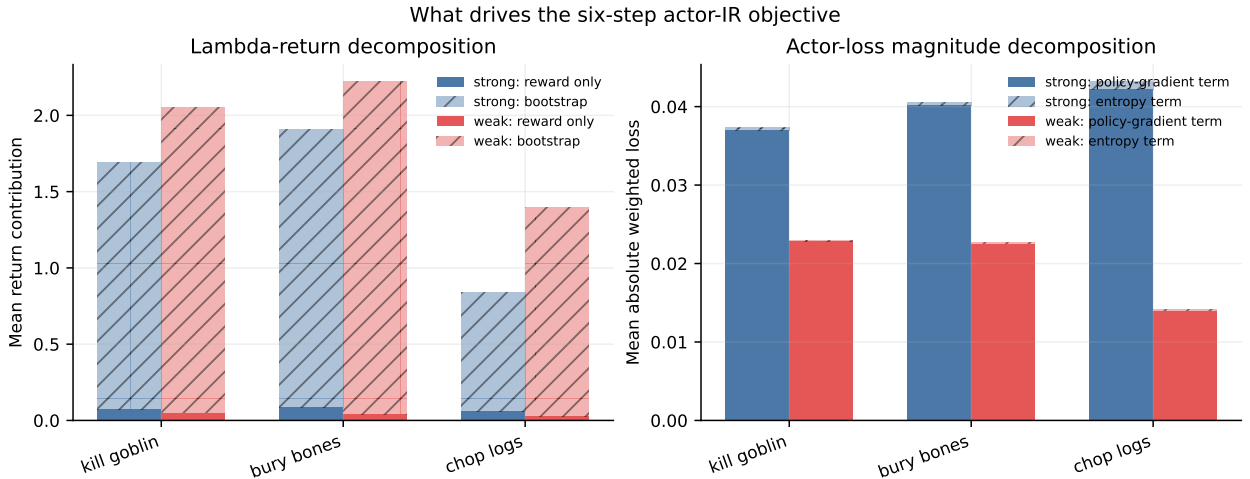


Figure 17: Study IIb decomposition of the exact six-step actor-IR objective. Left: lambda return is almost entirely critic bootstrap, and bootstrap grows at the weak checkpoint while imagined reward shrinks. Right: entropy is small relative to the policy-gradient term in measured scalar loss magnitude.

The weak model imagines fewer successes but larger returns. Factual imagined reward-event incidence falls from 6.76% to 3.52% of MCPB posterior samples, and reward-only return falls from 0.072 to 0.039. Nevertheless, bootstrap increases by 0.451 and total return increases by 0.419. Factual starting value rises from 1.315 to 1.668. The optimism observed on replay states is therefore present on the exact IR path and overwhelms the direction of the reward evidence.

Goal identity scarcely changes the teaching signal. Strong-checkpoint factual and counterfactual imagined returns are 1.450 and 1.447. At the weak checkpoint they are 1.868 and 1.865. On pre-completion anchors, factual versus wrong-goal reward-event rates are 34.0% versus 33.5% at 1.25M and 21.2% versus 20.8% at 1.5M. Requested-goal top-1 identification from imagined return degrades from 52.8% to 41.9%, and the factual-minus-wrong-goal mean margin is only 0.0029/0.0032.

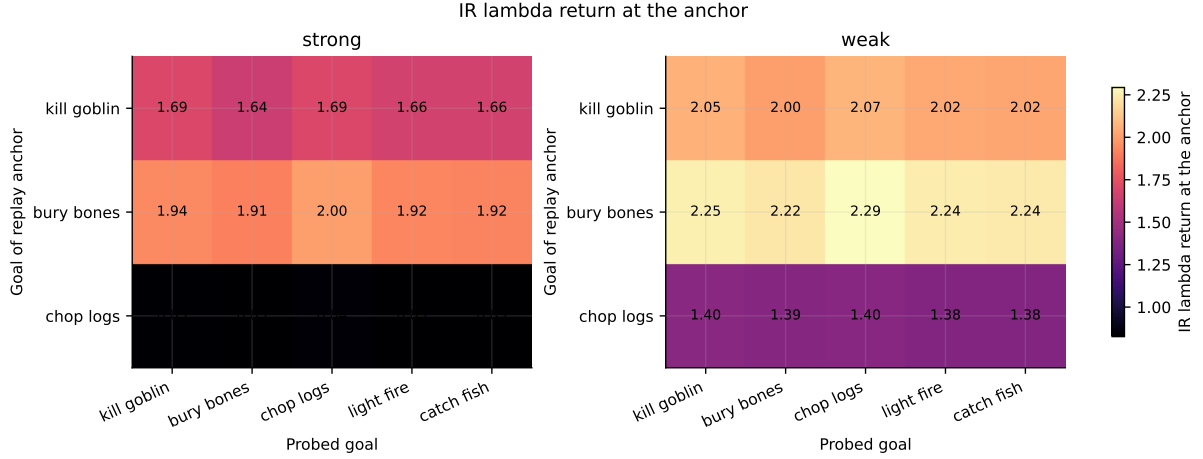


Figure 18: Mean IR lambda return by replay anchor goal (rows) and requested one-hot goal (columns). Row identity dominates column identity: the source trajectory matters far more than the goal requested for rehearsal. The weak checkpoint is uniformly more optimistic.

Reward and stopping events remain meaningful visual detectors: on successful trajectories, factual imagined reward-event incidence is 12.5% (strong) and 6.7% (weak), versus 1.8% and 0.7% on failed trajectories. Events concentrate near factual pre-completion states. The defect is not absence of all model signal; it is the missing same-state, wrong-goal distinction.

Entropy is not the dominant scalar term. Mean absolute factual policy-gradient loss is 0.0398 at 1.25M and 0.0196 at 1.5M. The corresponding entropy terms are 0.000552 and 0.000180, approximately 1.4% and 0.9% of those aggregate magnitudes; mean per-rollout ratios are 2.2% and 1.6%. These are scalar loss comparisons rather than gradient-norm comparisons, so an entropy ablation remains justified. They do not, however, support entropy as the primary explanation.

The actor itself becomes much more deterministic after collapse: factual policy entropy falls from 1.915 to 0.628. Simultaneously, the mean absolute policy-gradient term approximately halves. In other words, the damaged policy has less exploratory diversity while the learned objective intended to repair it becomes less informative and more bootstrap dominated.

8.5 Study II interpretation and resulting prediction

The two audits form a consistent mechanism. Reservoir replay preserved factual examples, so reward and continuation learned excellent visual event detection. It did not create the counterfactual labels in Equation 10; those heads can minimize factual loss while almost ignoring g . The critic then bootstraps on its own predictions through imagined states and develops values that are both too large and increasingly detached from realized success. Standard actor IR mostly optimizes this critic bootstrap.

This does not by itself prove the ultimate origin of critic optimism. Plausible contributors include self-reinforcing bootstrap targets, distribution shift into policy-imagined states, exploitation of model errors, weak goal selectivity, and continual drift in shared representations. What Study II establishes more narrowly is that *the critic is not a trustworthy teacher for the next actor-recovery experiment*. It therefore made a falsifiable prediction: if actor adaptation is mechanically viable, removing critic bootstrap should outperform standard IR; removing entropy alone should not.

9 Study III: controlled actor corruption and recovery

9.1 Question and experimental controls

Study III asks whether actor-only imagined adaptation can recover a deliberately damaged policy when every non-actor component is held fixed. It is a mechanism experiment, not another continual-training run. The source is the pinned 1.25M checkpoint, chosen because all three goals were simultaneously competent and Study II had already characterized its heads and imagination path.

The experiment applies an independent zero mask to actor parameters only. Encoder, RSSM, decoder, reward, continuation, online value, slow value, normalizers, and their optimizer states remain byte-identical to the source. Six mask probabilities, $p \in \{0.1, 0.2, 0.3, 0.4, 0.5, 0.6\}$, are calibrated on 20 sampled episodes per goal using paired reset seeds. The selection target is 35% aggregate retention of clean success, accepted only within 10%–65%, with no goal retaining more than 85% and a penalty for uneven or beneficial corruption.

The selected 40% zero mask affects all 11 actor parameter arrays (3,409,912 parameters) and has actor-relative L_2 distance 0.637 from the source, close to the $\sqrt{0.4}$ scale expected for independent masking. In calibration it retains 35.1% of aggregate clean success: per-goal retention is 50.0%, 20.0%, and 35.3% for kill, bury, and chop. The independent main seed schedule produces a less severe but still substantial sampled decline, from 56.3% clean mean to 28.3% corrupted mean.

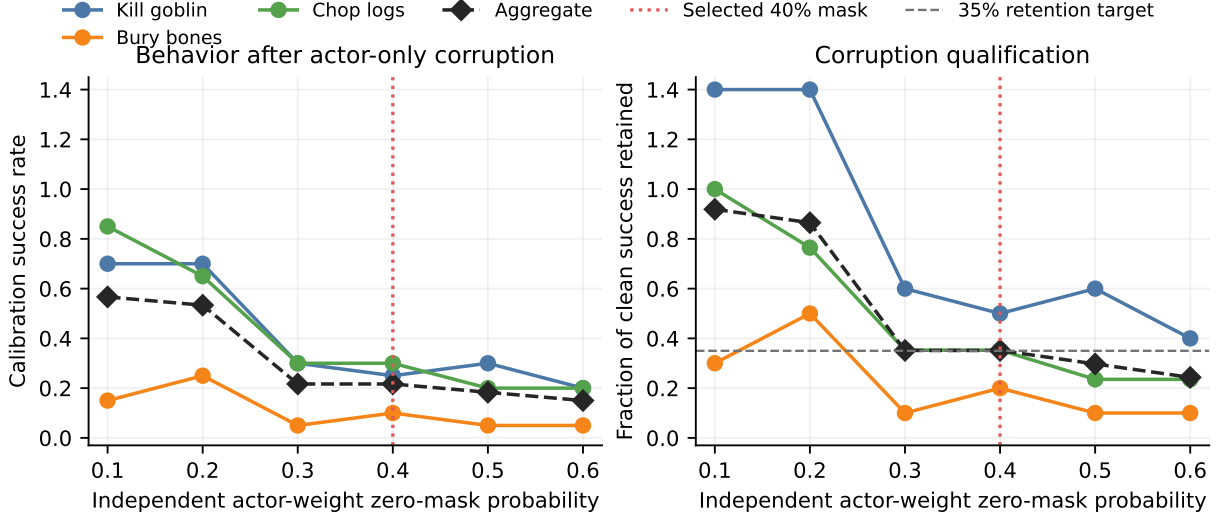


Figure 19: Study III corruption calibration. Six actor-only mask strengths were evaluated before the main assay. The 40% mask is the qualified candidate closest to the prespecified 35% aggregate-retention target while avoiding near-intact individual goals.

Every evaluation episode begins from the same persistent checkpoint parameters. Temporary actor parameters adapt from the first real posterior state and then once per environment action; they accumulate within the episode and are discarded at reset. Each update uses 128 posterior samples, learning rate 4×10^{-5} , and one optimizer step. The world model and both critics are frozen in every arm. Horizons are $H \in \{3, 6, 15, 20\}$.

9.2 Recovery objectives

Fifteen conditions separate three possible causes of failure.

No-adaptation controls. The clean and corrupted actors establish the attainable source behavior and the size of the induced deficit under the same main evaluation schedule.

Standard IR. This is the learned Dreamer actor objective used in Study I: policy gradients from normalized imagined lambda returns, including value bootstrap, plus entropy coefficient 3×10^{-4} . It is tested at all four horizons.

Entropy-off standard IR. This arm keeps the standard imagined return and critic bootstrap but sets the entropy coefficient to zero. It directly tests whether entropy, rather than the return target, caused harmful updates.

Reward-only IR. This arm uses predicted reward and continuation along the imagined rollout but sets terminal value bootstrap and entropy to zero. Abstractly,

$$J_{\text{RO}}(\theta; s, g) = \mathbb{E}_{p_\phi, \pi_\theta} \left[\sum_{k=0}^{H-1} \left(\prod_{j < k} \hat{c}_{t+j} \right) \gamma^k \hat{r}_{t+k} \right], \quad (12)$$

with the implementation retaining Dreamer’s differentiable policy-gradient machinery and giving posterior samples whose imagined trajectories contain zero reward no policy-gradient signal. This is a diagnostic critic-free objective, not an oracle: rewards and continuations are still learned model predictions.

Clean-reference distillation. The frozen clean actor supplies its full categorical action distribution on the current real posterior state. The temporary corrupted actor minimizes cross-entropy to that teacher. This horizon-independent positive control tests whether the corruption can be reversed under the same episode-local actor optimizer without relying on imagined returns.

9.3 Evaluation scale and integrity

Every condition is evaluated on all three goals with 100 sampled and 50 deterministic episodes. The main matrix therefore contains

$$15 \text{ conditions} \times 3 \text{ goals} \times (100 + 50) = 6,750 \text{ real-environment episodes.} \quad (13)$$

Calibration contributes 420 further episodes. In runner terms, all 21 calibration units and all 90 main units completed, giving 111 complete units with no failed attempt or retry. Goal, policy mode, and reset-seed identities match across conditions; the pairing audit reports no mismatch. Requested episode counts, reset counts, return codes, and persistent parameter-integrity digests all pass.

9.4 Study III results

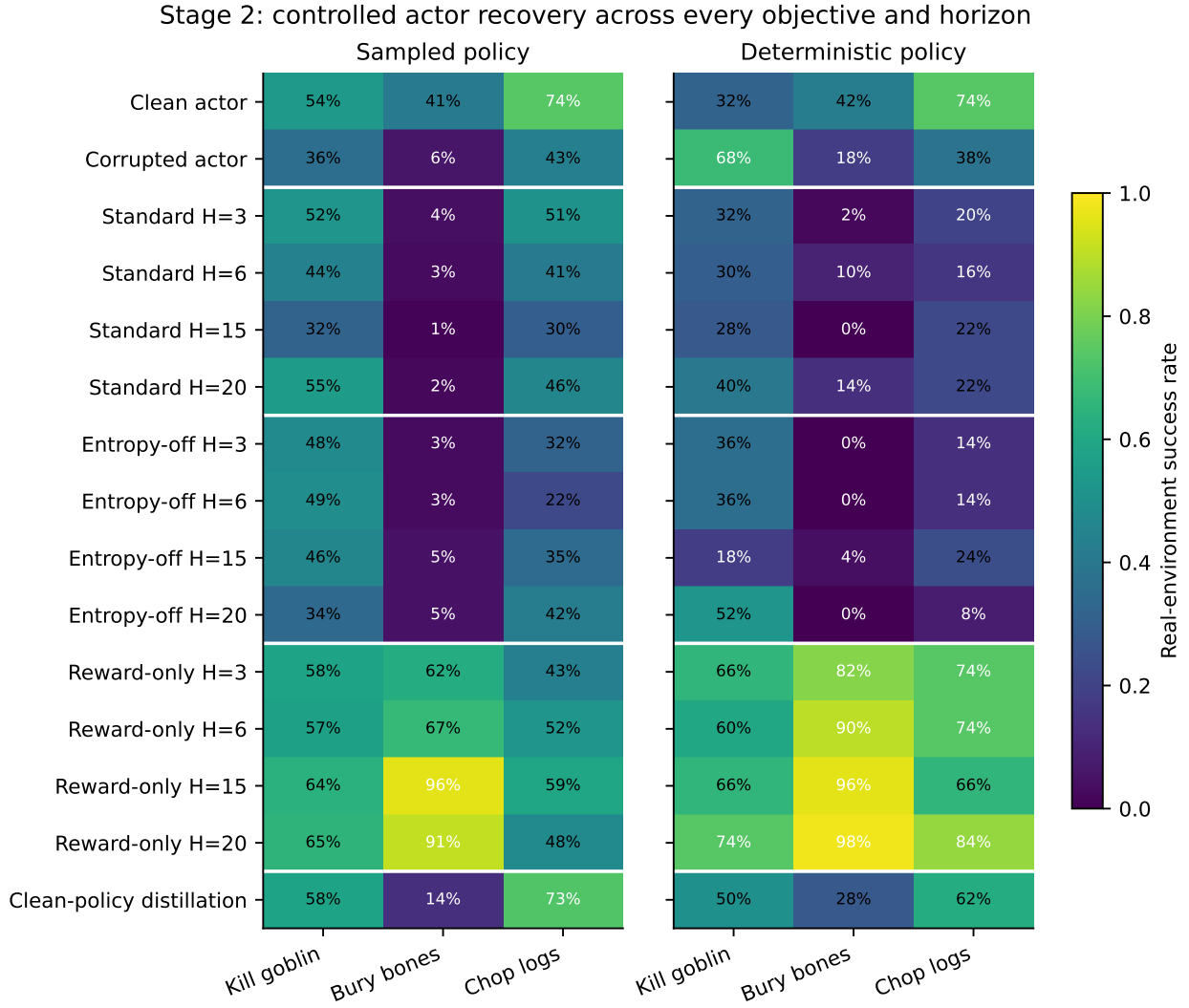


Figure 20: Complete Study III result matrix. Each cell is real-environment task success, not predicted reward. Standard and entropy-off IR remain weak, while reward-only IR recovers the actor and often exceeds the clean source behavior.

Table 3: Aggregate Study III success across all three goals. Sampled totals use 300 episodes per row; deterministic totals use 150. “Best” selects the horizon separately within each objective family and policy mode.

Condition	Sampled mean	Deterministic mean	Interpretation
Clean actor	.563	.493	Source checkpoint
Corrupted actor	.283	.413	Actor-only deficit
Standard IR, best	.357 ($H = 3$)	.253 ($H = 20$)	Weak sampled; destructive deterministic
Entropy-off IR, best	.287 ($H = 15$)	.200 ($H = 20$)	Removing entropy does not repair IR
Reward-only IR, $H = 3$	.543	.740	Near/full aggregate recovery
Reward-only IR, $H = 6$	.587	.747	Exceeds clean mean
Reward-only IR, $H = 15$	.730	.760	Best sampled result
Reward-only IR, $H = 20$	.680	.853	Best deterministic result
Clean-policy distillation	.483	.467	Partial positive-control recovery

Standard IR is narrow and unstable. At $H = 3$, sampled kill improves from 36% to 52% and chop from 43% to 51%, but bury remains near zero. Aggregate sampled success rises only to 35.7%. At $H = 6$ it falls to 29.3%; at $H = 15$ it drops below the corrupted baseline to 21.0%; at $H = 20$ it rebounds to 34.3%. Deterministic

success is damaged at every horizon, ranging from 16.7% to 25.3% versus 41.3% corrupted. Standard IR can discover a useful local direction for one policy/task combination, but its teaching signal is not robust.

Entropy removal does not solve the failure. Entropy-off sampled means remain between 24.7% and 28.7%; deterministic means remain between 15.3% and 20.0%. This agrees with Study IIb’s objective decomposition. Entropy may influence individual decisions, but it is neither necessary nor sufficient for the broad harm.

Removing critic bootstrap changes the outcome decisively. Reward-only IR reaches sampled means of 54.3%, 58.7%, 73.0%, and 68.0% as H increases from 3 to 20. Deterministic means are 74.0%, 74.7%, 76.0%, and 85.3%. The result is not merely recovery to the clean actor: the best reward-only conditions exceed it by 16.7 sampled points and 36.0 deterministic points.

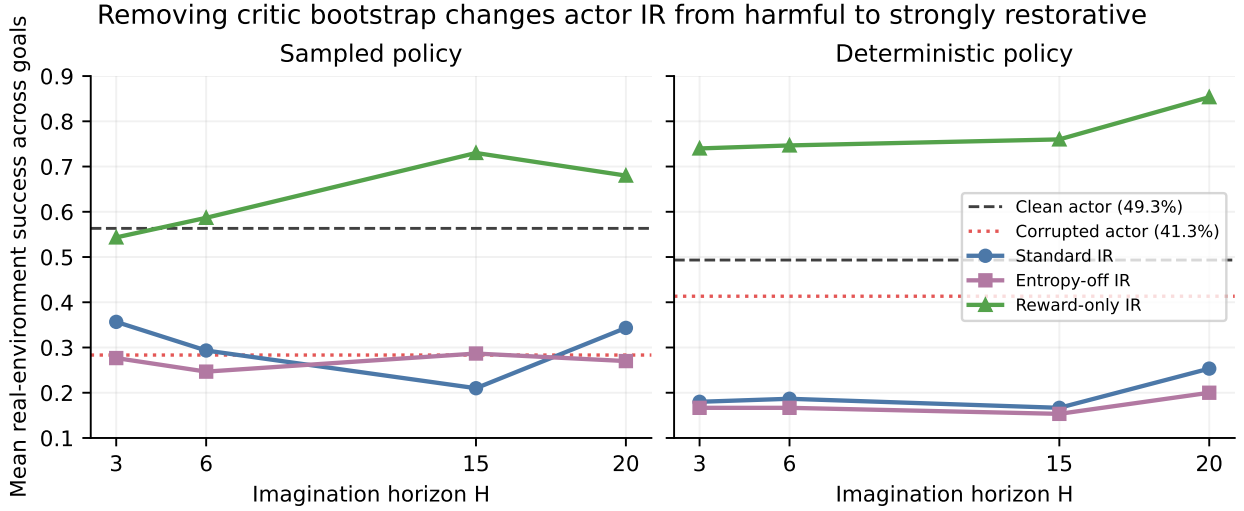


Figure 21: Aggregate horizon response. Standard and entropy-off IR track the corrupted baseline or fall below it. Reward-only IR crosses the clean-actor line and generally benefits from longer imagination. This is the central causal ablation of the chapter.

The per-goal pattern is informative. At sampled $H = 15$, reward-only IR obtains 64% kill, 96% bury, and 59% chop. Relative to paired corrupted episodes, changes are +28 points (95% interval [+15, +40]), +90 [+83, +96], and +16 [+2, +30]. All exclude zero. At $H = 20$, corresponding changes are +29 [+17, +41], +85 [+77, +92], and +5 [−8, +19]. Thus $H = 15$ is the most consistent sampled horizon; $H = 20$ is strongest under deterministic action selection.

Reward-only rehearsal recovers different skills at different horizon scales

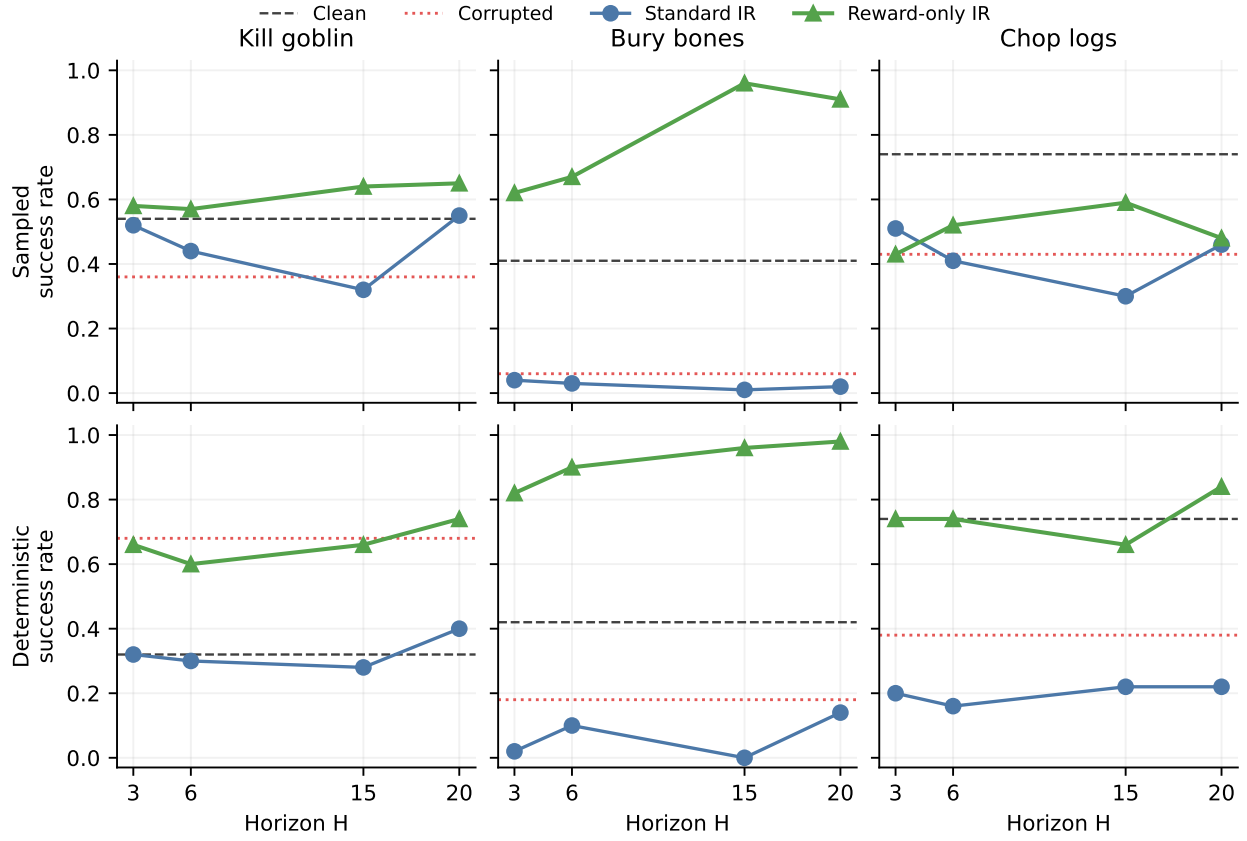


Figure 22: Goal-specific horizon response. Bury bones exhibits dramatic reward-only recovery, kill improves consistently under sampled control, and chop peaks earlier or weakens at sampled $H = 20$. There is no single horizon that dominates every task and policy mode.

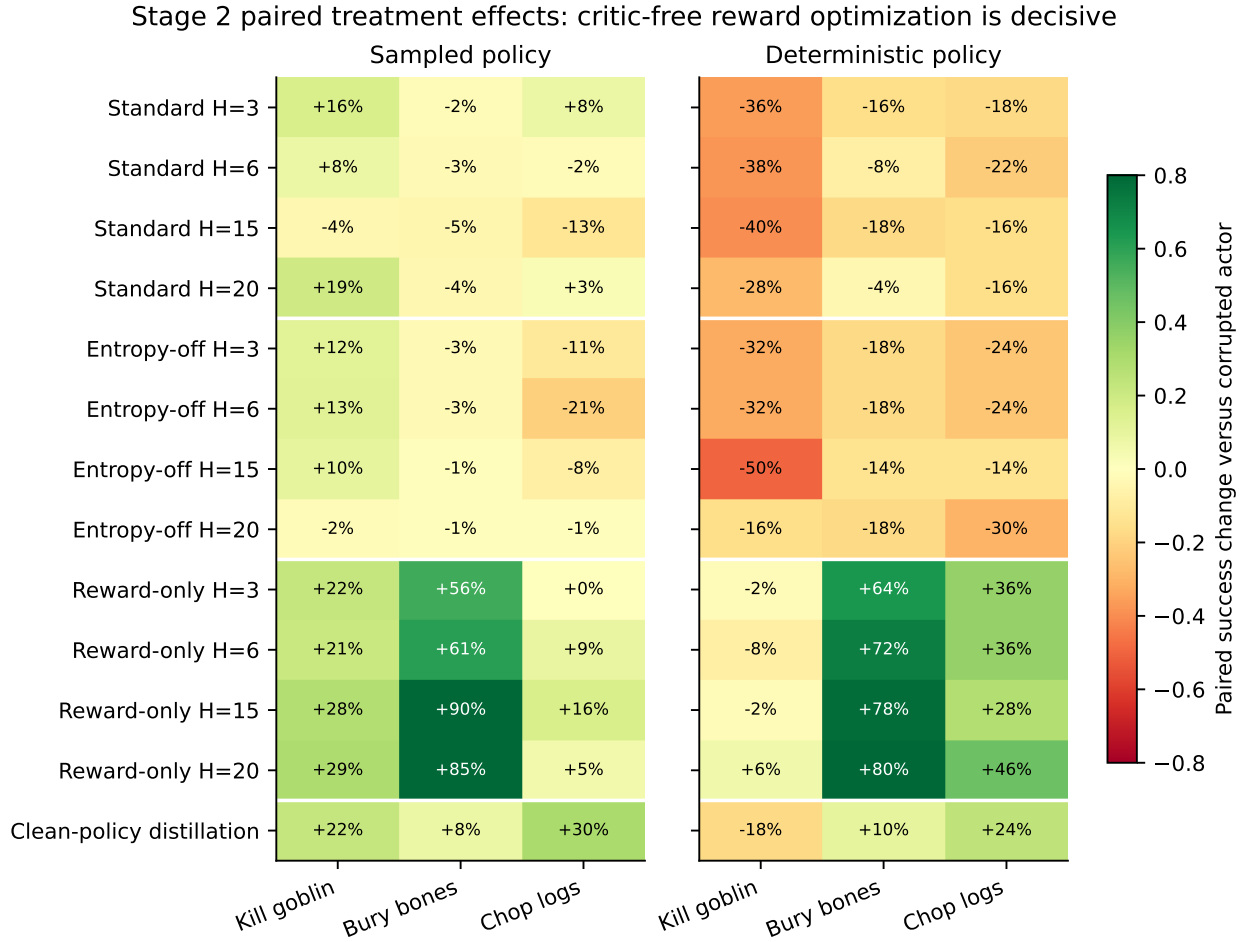


Figure 23: Paired changes relative to the corrupted actor. The green reward-only block contrasts sharply with standard and entropy-off objectives. Deterministic kill requires special caution because actor corruption itself improved its raw deterministic success from 32% clean to 68% corrupted.

Reference distillation provides partial mechanical recovery. Distillation raises sampled aggregate success from 28.3% to 48.3%: kill rises 36%→58%, bury 6%→14%, and chop 43%→73%. It does not exactly recreate the source policy within the limited episode-local updates, especially on bury, but it confirms that the optimizer can move the corrupted actor toward a trusted target. Reward-only IR’s much stronger bury result is therefore not explained by simple cloning of the original actor; it constitutes online policy improvement under the learned reward model.

9.5 Study III interpretation

Study III supports the prediction made before the run. The actor-adaptation machinery is functional; entropy is not the main obstacle; and critic bootstrap is the principal harmful component of the present IR objective. This is a stronger conclusion than a correlation between critic error and behavioral collapse because the assay intervenes on the objective while holding the checkpoint, corruption, posterior samples, optimizer schedule, environment seeds, and frozen parameters fixed.

The reward-only result also changes the interpretation of the learned world model. Study II showed that reward and continuation are not counterfactually goal selective at completion states, yet they still contain enough trajectory-level event structure to guide an actor to genuine real-environment success. In this arena, different goals require different visual/action paths even if the final event detector is shared. A goal-conditioned actor can exploit those reward gradients from a goal-specific starting context. This explains how reward-only IR can work without excusing the semantic defect: in a denser or more ambiguous task landscape, rewarding any completion-like state could become direct cross-goal interference.

Performance above the clean actor should be called *test-time policy improvement*, not memory recovery alone. It is encouraging because success is measured in RLScope rather than by predicted return. It still warrants tra-

jectory inspection: the model may have discovered a narrow but valid strategy, and horizon-specific degradation on chop suggests that compounding model error remains relevant.

10 Study IV: critic-free recovery from natural forgetting

10.1 Question and prospective design

Study III establishes a controlled possibility result: a retained world model can teach a synthetically damaged actor when the critic is removed from the objective. Study IV (canonical Stage 3B) asks the necessary external-validity question: does the same objective repair behavior lost *naturally* during sequential training? No actor corruption is applied. The inputs are the immutable 1.25M checkpoint, at which the three behaviors were relatively strong, and the naturally collapsed 1.5M checkpoint. The experiment was specified before launching any child evaluation and contains five conditions:

1. frozen checkpoint actor;
2. standard critic-bootstrapped IR at the historical $H = 6$ setting;
3. reward-only IR at $H \in \{6, 15, 20\}$.

All adapted conditions start from the unmodified checkpoint actor. Actor updates begin at the first real posterior state, occur once per environment action, accumulate within an episode, and are discarded at reset. Each update uses 128 Monte Carlo posterior samples and learning rate 4×10^{-5} . Standard IR retains the historical entropy coefficient 3×10^{-4} and Dreamer lambda-return bootstrap. Reward-only IR uses Equation 12: entropy and terminal critic bootstrap are zero, while the learned RSSM, reward, and continuation remain active. Encoder, RSSM, decoder, reward, continuation, online value, slow value, replay, normalizers, and persistent checkpoint state are frozen.

Every checkpoint-condition-goal cell is evaluated with 100 sampled and 50 deterministic episodes. The complete design therefore contains

$$2 \text{ checkpoints} \times 5 \text{ conditions} \times 3 \text{ goals} \times (100 + 50) = 4,500 \text{ episodes.} \quad (14)$$

Reset seeds are paired across the five conditions within checkpoint, goal, and policy mode. The primary outcome is real-environment success; secondary outcomes include paired discordance, actor update magnitude, action count, and imagined-compute dose.

Before the new evaluation, cells were labelled *weak* when their historical Stage I sampled success was below 50%. This labels 1.25M bury (historical 49%) and all three 1.5M goals as weak; 1.25M kill (56%) and chop (88%) are competent. The label is an oracle mechanism-analysis stratum, not a proposed deployment gate. The distinction matters because the independent Stage IV frozen schedule measured 1.25M bury at 71% sampled: the precommitted label remains historical, while all treatment effects correctly use the new paired frozen outcome.

10.2 Execution and integrity

The restart-safe workflow completed all 60 evaluation units on their first attempt in 23.60 hours. All 4,500 requested episodes and reset identities were present, all child processes returned successfully, and the pairing audit reported no mismatch. Persistent parameter-integrity digests were unchanged in every unit. The experiment used approximately 623,247 real actions. Adapted arms performed approximately 489,099 episode-local actor updates and 689 million imagined latent transitions; none of these imagined transitions is new environment experience.

10.3 Absolute recovery results

Figure 24 gives the primary sampled-policy matrix. At 1.25M, the frozen mean is 70.3%. Reward-only IR raises it at every horizon: 81.3% at $H = 6$, 83.7% at $H = 15$, and 87.0% at $H = 20$. Standard $H = 6$ IR instead reduces it to 40.3%. At 1.5M, the frozen mean is only 13.3%. Reward-only IR raises it to 21.7%, 39.7%, and 43.3%; standard IR reaches 19.0%.

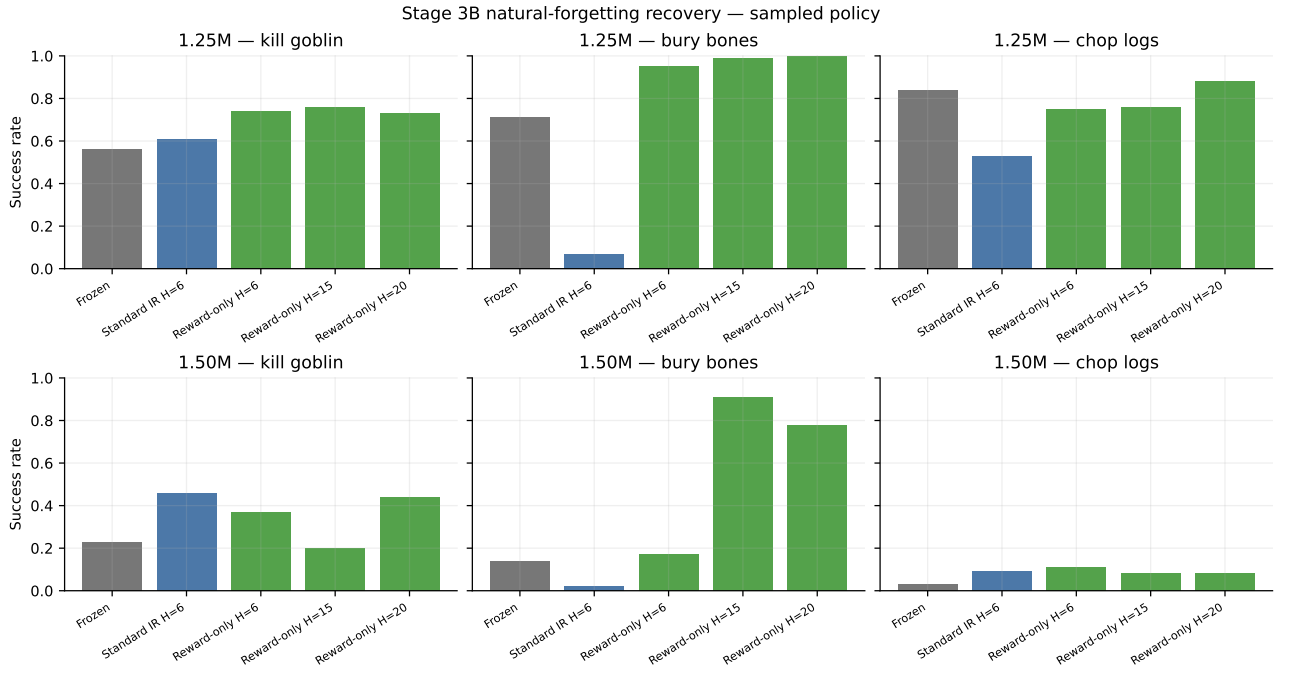


Figure 24: Study IV sampled-policy success (100 real-environment episodes per bar). Reward-only IR improves aggregate behavior at both checkpoints. The major natural recovery is 1.5M bury bones; standard IR catastrophically damages 1.25M bury and chop.

The deterministic evaluation reproduces the main result rather than merely reflecting exploratory action sampling. At 1.25M, reward-only $H = 20$ raises mean success from 59.3% to 84.7%; standard IR lowers it to 35.3%. At 1.5M, reward-only $H = 15$ is strongest, raising the mean from 11.3% to 42.7%. The different aggregate optimum— $H = 20$ for 1.25M but $H = 15$ for 1.5M deterministic control—is an early warning against a universal horizon rule.

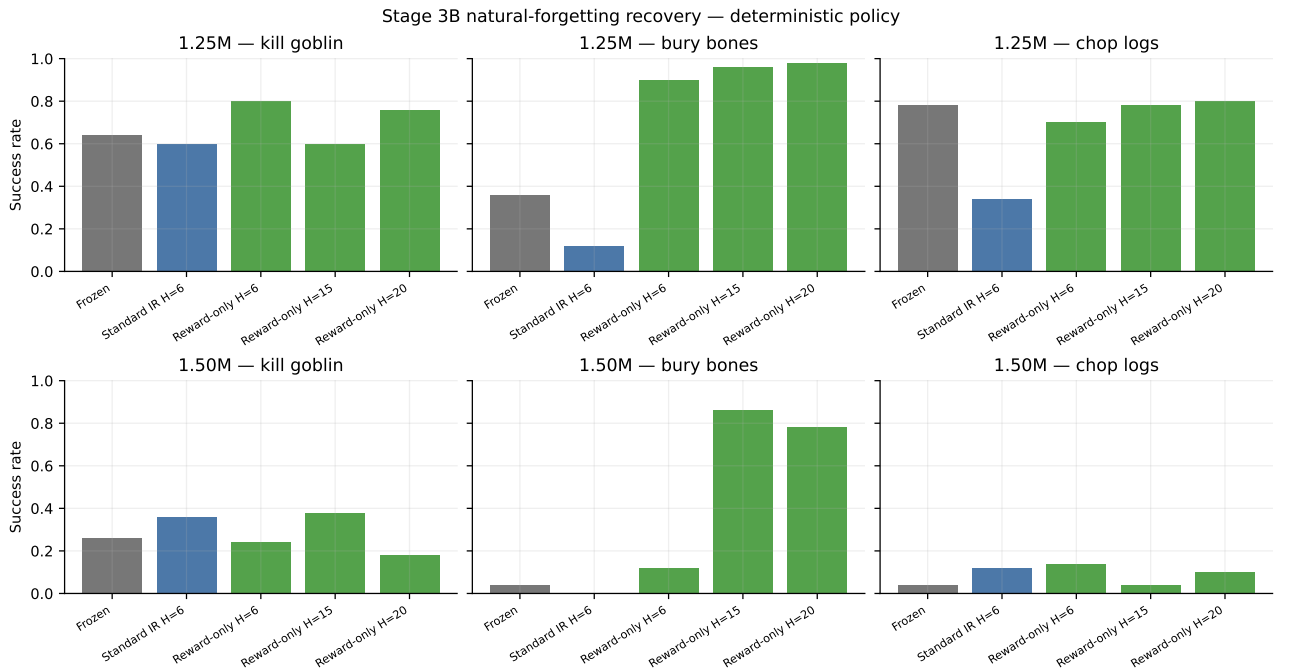


Figure 25: Study IV deterministic-policy success (50 episodes per bar). The bury recovery remains large without action sampling, while goal-specific horizon preferences become more apparent.

Table 4: Mean success across three goals in Study IV. Each sampled entry pools 300 episodes and each deterministic entry 150. Changes in parentheses are arithmetic mean changes from the paired frozen cells.

Checkpoint	Condition	Sampled	Deterministic
1.25M	Frozen	.703	.593
	Standard IR $H = 6$	.403 (−.300)	.353 (−.240)
	Reward-only $H = 6$	.813 (+.110)	.800 (+.207)
	Reward-only $H = 15$	.837 (+.133)	.780 (+.187)
	Reward-only $H = 20$	.870 (+.167)	.847 (+.253)
1.50M	Frozen	.133	.113
	Standard IR $H = 6$	.190 (+.057)	.160 (+.047)
	Reward-only $H = 6$	.217 (+.083)	.167 (+.053)
	Reward-only $H = 15$	.397 (+.263)	.427 (+.313)
	Reward-only $H = 20$	.433 (+.300)	.353 (+.240)

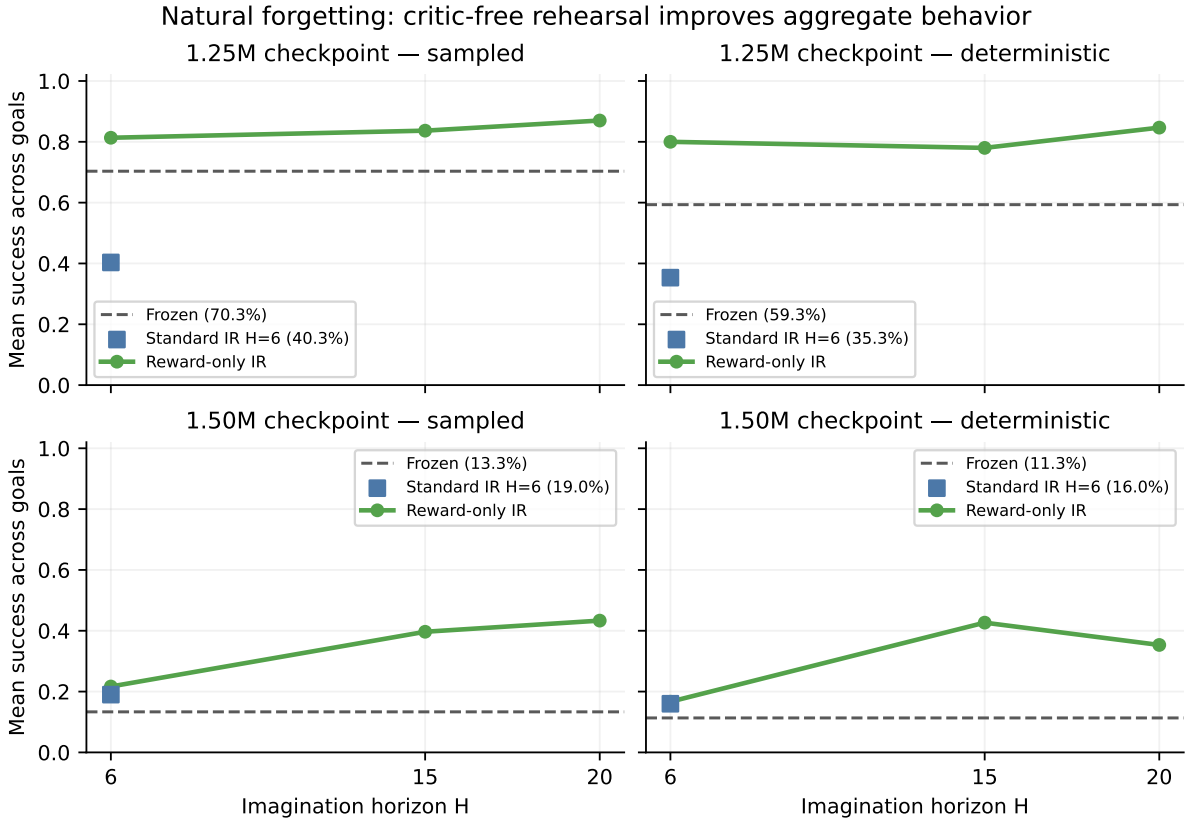


Figure 26: Aggregate horizon response on unmodified checkpoint actors. Reward-only IR exceeds the frozen mean in all twelve checkpoint-mode-horizon aggregates. Standard IR’s aggregate gain at 1.5M masks severe goal-specific asymmetry.

10.4 Paired effects and task-specific exceptions

The most compelling cell is naturally forgotten 1.5M bury bones. Reward-only $H = 15$ changes sampled success from 14% to 91%, a paired change of $+.77$ with 95% episode-bootstrap interval $+.68, +.85$. Deterministic success changes from 4% to 86%, $+.82[+.70, +.92]$. At $H = 20$ the corresponding changes remain large but are lower: $+.64[+.53, +.74]$ sampled and $+.74[+.62, +.86]$ deterministic. This is the first direct evidence in the programme that an imagined, critic-free test-time update can recover a severe natural forgetting deficit.

Other cells show both transfer and limits. At 1.25M, reward-only $H = 15$ improves sampled kill by $+.20$, with interval $+.08, +.32$. At $H = 20$, bury improves by $+.29$, with interval $+.20, +.38$. Strong chop is slightly damaged at shorter horizons: $-.09$ at $H = 6$ and $-.08$ at $H = 15$, before returning to $+.04$ at $H = 20$; all three chop intervals include zero. At 1.5M, $H = 20$ improves sampled kill by $+.21[+.09, +.33]$ but changes

deterministic kill by $-.08[-.26, +.10]$. Across the 12 prespecified weak sampled condition–horizon cells, reward-only IR improves 11 and harms one; across six competent cells it improves four and harms two.

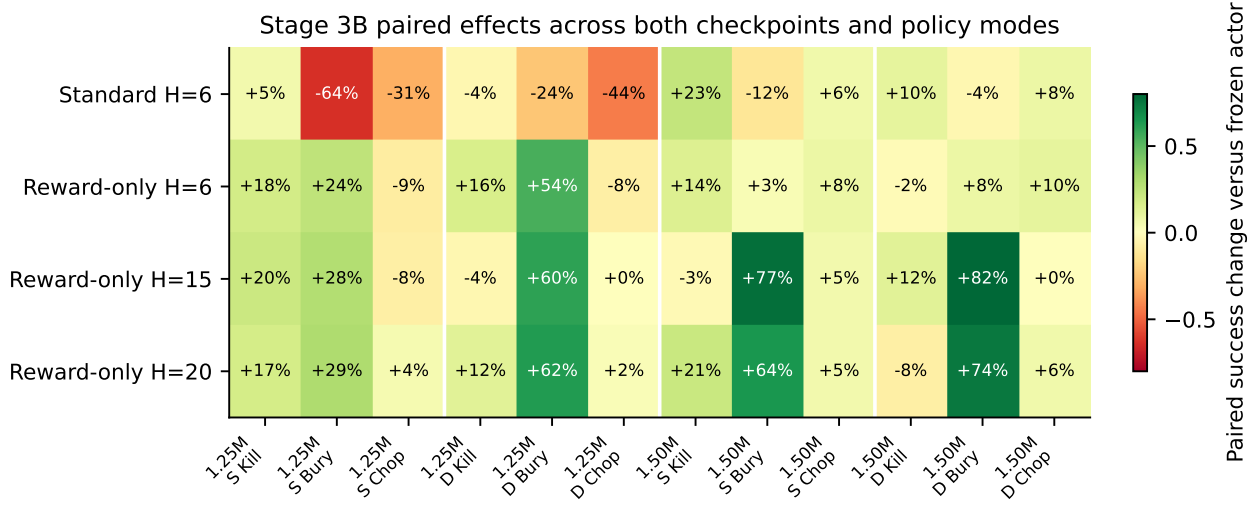


Figure 27: Complete Stage IV paired treatment-effect matrix. S and D denote sampled and deterministic policies. Critic-free recovery is broad but not uniform; the optimum depends on checkpoint, task, and action-selection mode.

Standard IR is conditionally useful but unsafe. It raises 1.5M sampled kill from 23% to 46%, $+.23[+.10, +.36]$, yet lowers 1.25M bury from 71% to 7%, $-.64[-.74, -.54]$, and chop from 84% to 53%, $-.31[-.42, -.20]$. A positive aggregate at the collapsed checkpoint therefore does not rehabilitate the critic objective; it demonstrates that the critic can occasionally point in a useful direction while remaining unsuitable for unconditional adaptation.

10.5 Study IV interpretation and claim boundary

Study IV supports its precommitted hypothesis. The Stage III result was not specific to an independent zero-mask perturbation: the same critic-free learned reward objective repairs at least one severe policy deficit created by real sequential training and improves aggregate behavior at both audited checkpoints. Because success is measured after the update in RLScapes rather than by predicted reward, the bury result is genuine behavioral recovery rather than merely an increase in model score.

The study also strengthens the causal account linking Studies II and III. Standard and reward-only arms share the checkpoint, actor optimizer, posterior starts, learned transition/reward/continuation models, environment protocol, and paired seed schedule. Removing the critic-dominated teaching signal reverses catastrophic 1.25M damage and enables large 1.5M recovery. Entropy is also absent in reward-only IR, but Study III’s entropy-off ablation already showed that entropy removal alone does not explain the reversal.

This remains episode-local test-time policy adaptation, not persistent continual learning. There is one training seed, no learned gate, and no guarantee that the best horizon can be known in advance. The prespecified weak label uses historical evaluation and is diagnostic privilege. Stage IV therefore establishes *where IR can work* and supplies target behavior for a later gate and trust region: retain the large weak-task gains without adapting destructively on already competent states.

11 Study V: critic provenance and calibration

Studies II–IV establish that the critic-dominated IR objective is harmful and that removing its bootstrap enables real behavioral recovery. They do not determine where the critic error originates. Canonical Stage 3A therefore decomposes four candidate explanations: value error already present on real posterior states; error accumulated as imagination depth increases; optimistic action sequences selected by the actor; and drift associated with weak goal conditioning. The audit is read-only and reuses the immutable 1.25M “strong” and 1.50M “weak” checkpoints.

11.1 Fixed state bank and estimands

The source is the same 384-episode Stage 1A state bank: 64 successful and 64 failed episodes for each trained goal, containing 48,299 factual states. From these episodes the audit extracts 1,433 valid posterior anchors and evaluates all five one-hot goal queries. For every anchor-goal pair it samples 128 Monte Carlo posterior trajectories to $H = 20$, then analytically reuses prefixes at $H \in \{1, 3, 6, 15, 20\}$. Actor-selected proposals are compared with recorded-action proposals from the same posterior anchors whenever sufficient future actions exist.

For a factual replay state, critic calibration uses the realized discounted reward remaining in its recorded episode as the target. Counterfactual goals and actor-selected imagined trajectories are not assigned false oracle returns. Their role is comparative: goal queries are contrasted at the same state, and actor proposals are contrasted with recorded actions under the same posterior start. For each prefix the exact IR teaching signal is decomposed into

$$G_H^{\text{IR}} = G_H^{\text{reward}} + G_H^{\text{bootstrap}}, \quad (15)$$

alongside continuation, online value, slow value, entropy, and imagined-state diagnostics. Neither agent parameters nor optimizer, replay, or checkpoint state are updated.

Both audits completed with identical before/after parameter digests. Each checkpoint contains 35,825 actor-prefix and 26,365 recorded-prefix summaries, or 62,190 rows; the combined audit contains 124,380 summaries. The actor proposal alone represents 7,165 anchor-goal rollouts and 18,342,400 imagined latent transitions per checkpoint.

11.2 The critic is already wrong on real posterior states

Figure 28 and Table 5 give the central result. The two checkpoints see exactly the same 48,299 factual states and targets, whose state-weighted mean is 0.176. The online critic instead predicts 1.359 at 1.25M and 1.771 at 1.50M. Because these episodes contain at most one sparse completion reward, values above one are structurally overoptimistic rather than merely conservatively high.

Table 5: Real-posterior critic calibration on the identical factual state bank. Correlations are unweighted means over the three trained goals; target, prediction, bias, and MAE are state-count weighted.

Checkpoint	Target	Prediction	Bias	MAE	Pearson	Spearman
1.25M strong	.176	1.359	+1.184	1.192	.198	.213
1.50M weak	.176	1.771	+1.595	1.595	.058	.081

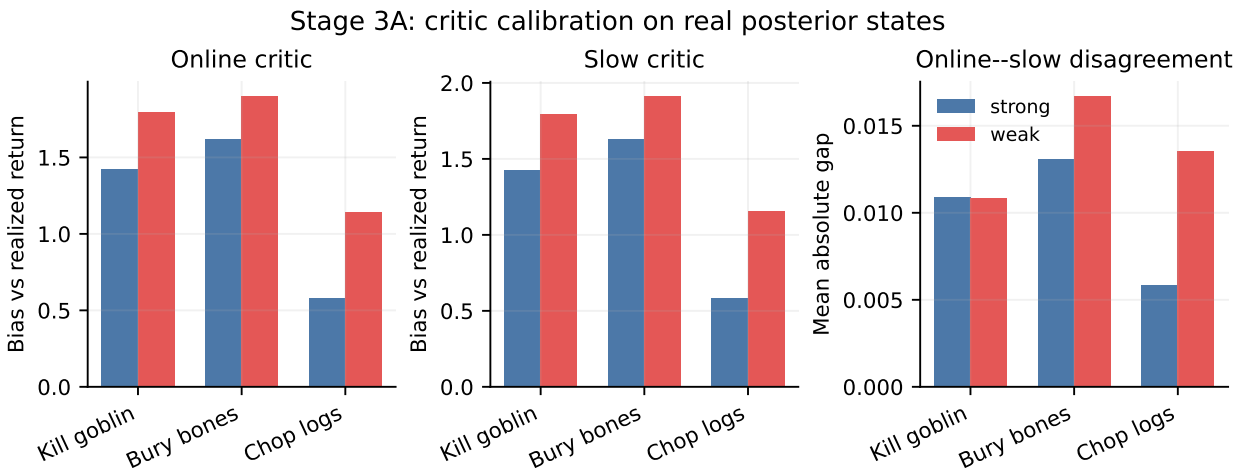


Figure 28: Critic calibration on real posterior states. Bias grows by 0.412 (34.8%) at the collapsed checkpoint while value ranking nearly disappears. The slow critic closely tracks the online critic rather than providing an independent check.

The failure is not only a common intercept error. Goal-level biases at 1.25M are +1.421 for kill, +1.621 for bury, and +0.579 for chop; at 1.50M they rise to +1.793, +1.900, and +1.142. Mean Pearson correlation falls from

.198 to .058 and Spearman from .213 to .081. The critic therefore loses both calibration and useful ordering as behavior collapses. A scalar offset correction would leave an extremely weak policy-improvement signal.

The slow critic does not solve this problem. Its weighted biases are +1.190 and +1.603, and its mean absolute prediction gap from the online critic is only .0098 and .0136. It is a delayed stabilizer for the same learned target, not an independently calibrated evaluator or uncertainty estimate.

11.3 Horizon decomposition: no late runaway

If open-loop model error were progressively inflating the target, longer prefixes should produce continuing growth. The observed pattern is different. Table 6 and Figure 29 show that the factual actor-selected target is already almost entirely bootstrap at $H = 1$ and plateaus by $H = 3$ –6. From $H = 6$ to $H = 20$, the strong full target changes from 1.480 to 1.482 and the weak target from 1.891 to 1.891.

Table 6: Actor-selected factual IR target decomposition. Values are unweighted means over the three trained goals, matching the plotted estimand.

Checkpoint	H	Full target	Reward-only	Bootstrap	Bootstrap fraction
1.25M	1	1.443	.052	1.391	95.9%
1.25M	6	1.480	.072	1.407	94.4%
1.25M	20	1.482	.106	1.376	91.7%
1.50M	1	1.848	.036	1.812	97.5%
1.50M	6	1.891	.039	1.852	97.4%
1.50M	20	1.891	.050	1.841	96.9%

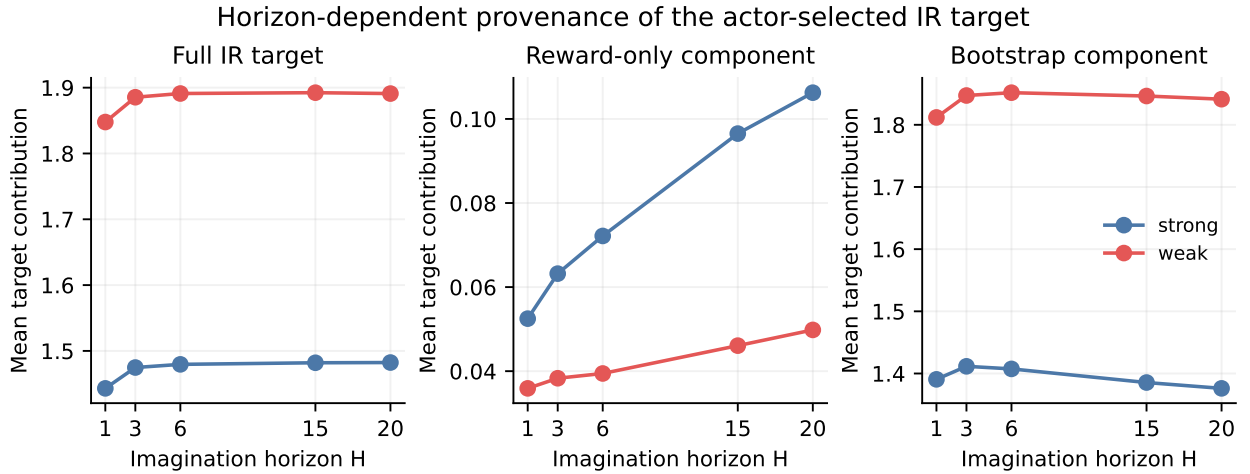


Figure 29: IR target provenance across horizon. Longer imagination does not create a $H = 6$ to $H = 20$ explosion. The collapsed checkpoint imagines less goal reward but assigns substantially more bootstrap value.

At 1.25M, added depth increases predicted reward while bootstrap slightly falls. At 1.50M, both components are nearly flat after $H = 3$. Crucially, the weak checkpoint predicts *less* reward than the strong checkpoint while assigning much larger total return. This locates the checkpoint difference in value bootstrap rather than accumulating reward-model hallucinations. Recorded-action proposals are likewise optimistic: their mean return bias spans +1.17 to +1.37 at 1.25M and +1.50 to +1.73 at 1.50M across the tested horizons. Actor exploitation is therefore unnecessary for the error to exist.

11.4 Actor selection amplifies the critic error

Although the actor does not create the critic’s bad scale, it preferentially selects trajectories with still higher bootstrap values. Relative to paired recorded actions, the actor raises the full target at every checkpoint and horizon (Table 7). At the collapsed checkpoint the apparent advantage comes entirely from bootstrap while predicted reward is lower.

Table 7: Actor-minus-recorded factual proposal differences from the same posterior anchors. Positive bootstrap and negative reward identify critic-seeking rather than predicted task progress.

Checkpoint	H	Full target	Bootstrap	Reward-only
1.25M	1	+.139	+.211	-.072
1.25M	6	+.126	+.138	-.012
1.25M	20	+.142	+.129	+.013
1.50M	1	+.212	+.298	-.086
1.50M	6	+.175	+.194	-.019
1.50M	20	+.182	+.196	-.015

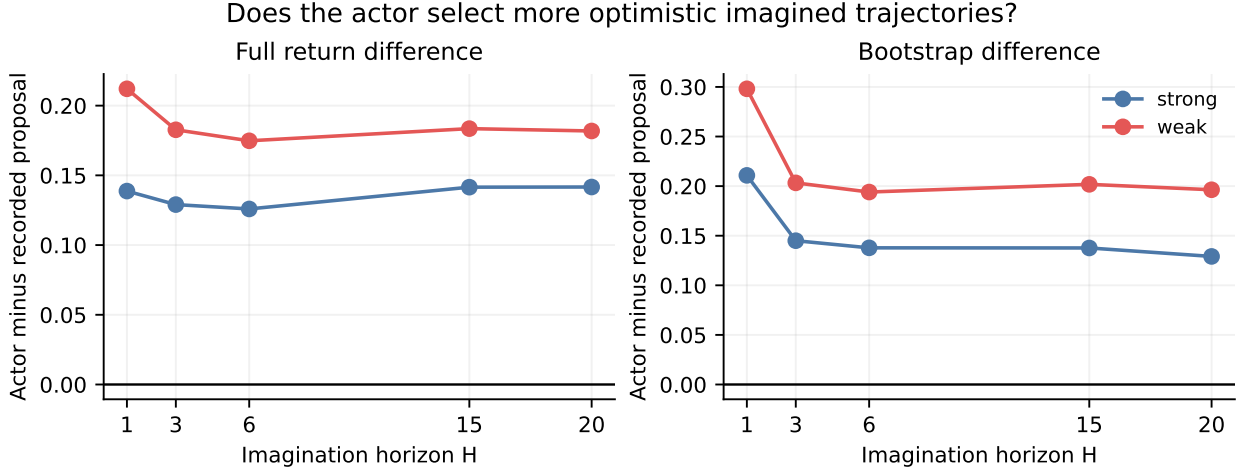


Figure 30: Actor-selected minus recorded-action imagined targets. The actor consistently selects higher-value prefixes; at 1.50M this occurs despite lower predicted reward. Model/critic exploitation is real, but it adds roughly 0.18 target units atop real-state bias of roughly 1.60.

The signature is clearest on bury bones. At $H = 6$, actor-minus-recorded bootstrap is +.266 at 1.25M and +.289 at 1.50M, while reward differences are $-.036$ and $-.043$. Thus the actor has learned to navigate the critic’s optimistic landscape rather than the reward model’s completion landscape. The effect is a secondary amplifier: it adds approximately .13–.14 full-return units at the strong checkpoint and .17–.21 at the weak checkpoint, compared with real-state biases already exceeding one.

Finally, actor-selected factual and pooled counterfactual targets remain nearly indistinguishable. Their mean full-target difference is below .001 at $H = 6$ and .008 at $H = 20$ at both checkpoints; reward-only differences remain below .005 at the headline horizons. This agrees with the direct Study IIa confusion matrix: checkpoint identity and value scale dominate the requested one-hot goal. Because Stage V pools four alternative goal queries, this observation supports but does not replace Study IIa’s stronger same-state counterfactual test.

11.5 Study V interpretation and claim boundary

Stage V refines the earlier “model exploitation” hypothesis. Exploitation is present, but it is not the origin of the optimism. The primary failure is a structurally overoptimistic, poorly ranked critic on real posterior states; the continual checkpoint change worsens that landscape; and the actor then selects trajectories that exploit its bootstrap. Increasing horizon from six to twenty does not cause runaway target growth, and the slow critic supplies no independent protection.

This is a provenance audit, not a unique causal diagnosis of why critic training drifted. Candidate upstream causes include recursive bootstrapping on biased targets, insufficiently goal-discriminative rewards and values, distribution shift across sequential phases, and the absence of task-semantic bounds on a success-return value. The audit nevertheless rules out three tempting remedies: simply shorten the existing critic-backed horizon, replace the online critic with its slow copy, or subtract one global calibration offset. Any future critic reuse should first demonstrate bounded calibration and useful ranking on held-out real states.

12 Cross-study synthesis

The experiments support the following chain of evidence:

1. Reservoir replay retains old observations and can temporarily recover all three behaviors, but balanced availability does not stabilize continual optimization.
2. Factual completion detection survives collapse, so failure is not wholesale erasure of reward examples.
3. Factual-only replay does not teach the same latent completion state to mean different things under different goals; reward and continuation largely ignore the requested goal in counterfactual queries.
4. The critic is badly overoptimistic, loses ranking ability, and grows more optimistic after real behavioral collapse.
5. The standard six-step IR return is 94%–97% critic bootstrap, so actor updates mostly inherit critic error rather than predicted task reward.
6. Entropy is small in the audited loss and removing it does not rescue controlled recovery.
7. Removing critic bootstrap produces large, paired, real-environment recovery. IR is therefore viable, but the current Dreamer value target is not a safe test-time teacher.
8. Critic-free recovery transfers from artificial actor masking to unmodified continual checkpoints: it improves all twelve checkpoint–mode–horizon aggregate comparisons and nearly restores naturally forgotten 1.5M bury behavior.
9. The transfer is not uniform. Strong chop can be damaged, and the best horizon changes by checkpoint and policy mode; a deployable method therefore requires a gate and conservative update constraint rather than unconditional optimization.
10. Critic error is already severe on real posterior states and worsens on an identical state bank after continual collapse; imagination depth is not required to produce it.
11. Actor-selected imagination is a secondary amplifier: it chooses higher-bootstrap trajectories that often have lower predicted reward than paired recorded actions.
12. The slow critic closely reproduces the online critic’s scale and error, so online–slow agreement cannot serve as a safety gate in its current form.

The initial hypothesis—that model exploitation slowly inflates critic targets during continual learning—is therefore only partly correct. Actor exploitation is measured directly, but the larger error precedes it: the critic is already badly calibrated at real posterior states. Nor does value grow progressively from $H = 6$ to $H = 20$. The remaining upstream question concerns critic learning itself: recursive target bias, goal-semantic aliasing, continual distribution shift, or their interaction. Longer continual runs may amplify these mechanisms, so bounded value targets and explicit counterfactual supervision are likely to become more important rather than less.

13 Revised experimental ladder

The chronology changes the priorities proposed after Study I.

1. **Stage 4A: counterfactual and bounded head repair.** Before another continual run, freeze the retained representation and actor at a pinned checkpoint and retrain diagnostic reward/value heads from preserved experience. Generate semantically valid targets for every goal, constrain success-return values to their attainable range, and evaluate on held-out real states. Repeat the Study IIa and Study V audits before allowing the repaired value into IR.
2. **Stage 4B: conservative critic reintroduction.** Compare the established reward-only teacher against a calibrated bounded bootstrap, clipped advantage, uncertainty penalty, short trusted bootstrap, or $J = J_{\text{RO}} + \beta J_{\text{value}}$ with small or state-gated β . Stage IV shows that the critic is occasionally useful, so complete removal need not be the final algorithm; Stage V shows that the present critic must not be reused unchanged.

3. **Stage 5: prespecified IR gate and trust region.** Use Stage IV only to nominate signals; freeze the decision rule before evaluation. The gate must detect when adaptation is needed without privileged historical success, while an action-KL or parameter-update constraint limits harm. Evaluation should report recovered weak cells, damaged competent cells, and compute avoided.
4. **Replay-mixture continual experiment.** Preserve Algorithm R retention but train primarily on the current task plus a controlled old-goal reservoir mixture, with update dose matched where possible. Compare gated reward-only/conservative IR against a no-IR baseline at pinned checkpoints.
5. **Stage 6: full validation.** Replicate across training seeds and task orders, then extend from three to five RLScape goals. Add return phases only after the mechanism is stable, to quantify relearning speed, backward transfer, and real-sample savings.

14 Global limitations

The combined evidence is unusually deep for one training trajectory but remains exploratory.

- **One continual-training seed.** Episode-level paired intervals do not quantify variation across independently trained world models and critics.
- **Adaptive research sequence.** Studies II and III were designed after inspecting Study I, Study IV after Study III, and Study V after the critic-free recovery result. Each workflow was specified before its child evaluations, but the overall sequence remains mechanism discovery rather than an independent confirmatory programme.
- **Corruption calibration and deterministic anomaly.** The mask was selected on sampled episodes. Under the independent main schedule, corruption improves deterministic kill success from 32% to 68%; aggregate deterministic “recovery fractions” are therefore not meaningful for that goal. Raw success and paired changes are reported instead.
- **Reward-only is model based, not oracle based.** It removes the critic but still trusts learned reward, continuation, and transition predictions. Success in the real environment validates outcomes, not calibration everywhere in latent space.
- **No persistent learning in Studies III–IV.** Adaptation resets each episode. The studies establish test-time recovery, not yet a continual-learning update rule that safely carries knowledge forward.
- **Historical weak labels are not gates.** Study IV fixed weak/competent status from the earlier sampled evaluation. The independent frozen schedule shifted some absolute rates, most visibly 1.25M bury from historical 49% to 71%. Treatment effects use paired contemporary baselines, but the label itself cannot be deployed online.
- **Stage V is diagnostic rather than interventional.** Its same-state design localizes critic error and actor amplification, but does not uniquely identify whether recursive bootstrapping, weak goal semantics, sequential distribution shift, or another optimizer interaction caused critic drift.
- **No oracle return for imagined counterfactuals.** Realized returns calibrate factual recorded states. Actor-selected and alternative-goal paths are assessed through paired contrasts, not falsely labelled with unobserved ground truth.
- **Shared simple action interface.** All tasks use a 504-way left-click grid and the same compact arena. Results may change with harder RLScape tasks, longer temporal dependencies, or physical robotic control.
- **Objective comparisons are not compute neutral.** Longer horizons require more imagined transitions and reward-only conditions often finish episodes sooner. Scientific comparison should report both behavioral gain and test-time compute.

15 Chapter conclusion

The programme began with a replay hypothesis: preserve older goals in a reservoir so that goal-conditioned Dreamer heads remain useful for IR. Reservoir replay did preserve information and yielded striking intermediate multi-goal competence, but it did not prevent synchronized collapse. Direct audits then revealed why ordinary

IR was unreliable. The reward and continuation heads recognize completion-like events but lack counterfactual goal semantics; the critic is severely optimistic; and almost the entire standard IR target is critic bootstrap.

Controlled actor corruption turns that diagnosis into a causal result. Standard IR and entropy-off IR do not reliably recover the actor. Reward-only IR does, reaching 73.0% sampled and 85.3% deterministic mean success and exceeding the original clean actor under the same evaluation schedules. Study IV then removes the artificial corruption: on naturally damaged actors, reward-only IR improves aggregate success at every tested horizon and recovers 1.5M bury from 14% to 91% sampled and 4% to 86% deterministic. Ordinary IR still catastrophically damages competent 1.25M behaviors.

Study V explains the failed teacher more precisely. On identical real states, critic bias grows from +1.184 to +1.595 as ranking collapses; the slow critic mirrors the error; target scale plateaus by short horizons; and the actor selects additional high-bootstrap, often lower-reward trajectories. The research direction therefore remains viable, but its center of gravity changes: the immediate opportunity is not more ungated Dreamer actor updates. It is **critic-free or conservatively critic-gated imagined adaptation, supported by bounded value learning, explicit counterfactual goal training, and replay that preserves old experience without overwhelming current-task learning.**

This is a positive result in the intended scientific sense. It identifies a condition under which IR works strongly on both synthetic and natural policy deficits, identifies and localizes the learned component that makes the default method fail, exposes the task-specific harm that still requires gating, and leaves a checkpointed, reproducible ladder for converting episode-local recovery into a continual reinforcement-learning algorithm. The next constructive study should repair goal semantics and value calibration at a pinned checkpoint before another costly continual-training run.

A Complete frozen-policy evaluation table

Table 8: Frozen success by checkpoint. S=sampled (100 episodes); D=deterministic (50 episodes). Dash means the goal had not yet been introduced.

Replay	Goal/mode	250k	500k	750k	1.0M	1.25M	1.5M
FIFO	Kill S	.91	.96	.75	.06	.98	.73
	Kill D	.80	.94	.70	.04	.92	.76
	Bury S	–	–	.21	1.00	.08	.12
	Bury D	–	–	.30	1.00	.28	.04
	Chop S	–	–	–	–	1.00	1.00
	Chop D	–	–	–	–	1.00	1.00
Reservoir	Kill S	.49	.78	.70	.02	.56	.16
	Kill D	.26	.74	.76	.02	.46	.24
	Bury S	–	–	.51	.04	.49	.08
	Bury D	–	–	.48	.00	.34	.04
	Chop S	–	–	–	–	.88	.05
	Chop D	–	–	–	–	.60	.06

B Reservoir IR effects and paired intervals

Table 9: Actor IR minus frozen success. Intervals are episode-order paired bootstrap 95% intervals; S uses 100 episodes and D uses 50. These intervals do not capture training-seed uncertainty.

Checkpoint	Goal	Mode	Change	95% interval
250k	Kill	S	−.13	[−.25, .00]
250k	Kill	D	+.02	[−.10, .14]
500k	Kill	S	−.05	[−.16, .06]
500k	Kill	D	+.04	[−.12, .20]
750k	Kill	S	−.02	[−.12, .08]
750k	Kill	D	−.10	[−.22, .02]
750k	Bury	S	−.22	[−.33, −.10]
750k	Bury	D	−.12	[−.32, .08]
1.0M	Kill	S	+.01	[−.02, .05]
1.0M	Kill	D	+.02	[.00, .06]
1.0M	Bury	S	−.03	[−.08, .01]
1.0M	Bury	D	.00	[.00, .00]
1.25M	Kill	S	+.03	[−.10, .16]
1.25M	Kill	D	+.10	[−.08, .28]
1.25M	Bury	S	−.18	[−.30, −.06]
1.25M	Bury	D	−.18	[−.34, −.02]
1.25M	Chop	S	−.36	[−.47, −.25]
1.25M	Chop	D	−.18	[−.36, .00]
1.5M	Kill	S	+.12	[.02, .23]
1.5M	Kill	D	+.16	[−.04, .36]
1.5M	Bury	S	−.02	[−.09, .05]
1.5M	Bury	D	−.02	[−.08, .04]
1.5M	Chop	S	+.07	[.00, .14]
1.5M	Chop	D	+.02	[−.08, .12]

C Reproducibility manifest

Table 10: Key configuration recorded in the reservoir experiment manifest.

Item	Value
Git commit	<code>ae2e1d25ed5eddc1289dddf36a946e24242d0c22</code>
RLScape	0.1.6; 28×18 click grid; 200-action limit; sparse success reward
Curriculum	kill goblin $\rightarrow$ bury bones $\rightarrow$ chop logs; 500k steps each; seed 0
Dreamer	“50m” preset; batch 8; sequence 32; train ratio 32; imagination horizon 15; bfloat16
Goal conditioning	Five-way one-hot; reward, continuation, actor, value, slow value conditioned; RSSM state goal-agnostic
Reservoir	1,000 complete episodes per goal; Algorithm R; uniform goal $\rightarrow$ episode $\rightarrow$ sequence sampling; persistent across phases
Checkpoints	250k, 500k, 750k, 1.0M, 1.25M, 1.5M; all pinned
IR	Actor only; every action; one update; $H = 6$; 128 posterior samples; $\alpha = 4 \times 10^{-5}$; entropy 3×10^{-4} ; critic/world model frozen
Evaluation	100 sampled and 50 deterministic episodes per condition; frozen/IR seed schedules paired

The report directory includes `generate_figures.py` and compact CSV extracts for the evaluation matrix, 50k training bins, and resource milestones. Running

```
MPLCONFIGDIR=/tmp/rlscape-report-mpl python generate_figures.py
latexmk -pdf report.tex
```

regenerates the figures and paper from the compact audited summaries.

D Study II–V reproducibility manifest

Table 11: Key configuration for the diagnostic audits and two recovery assays.

Item	Value
Study II checkpoints	Strong 1.25M and weak 1.5M milestones from the reservoir run
Study II state bank	384 episodes: 64 successful and 64 failed per trained goal; 48,299 states; source bank pinned at 1.25M
Head sweep	Reward, continuation, online value, slow value under all five one-hot goal IDs
Imagination audit	1,433 valid anchors; five-goal sweep; 128 paired posterior samples; $H = 6$; 7,165 rollouts and 5,502,720 imagined transitions per checkpoint
Study III source	1.25M checkpoint; actor-only independent zero mask; seed 0; selected strength 0.4
Study III objectives	Standard lambda-return IR; entropy-off standard IR; reward-only IR; clean-reference policy distillation
Study III horizons	$H \in \{3, 6, 15, 20\}$; one actor update per real action; 128 posterior samples; $\alpha = 4 \times 10^{-5}$
Frozen components	Encoder, RSSM, decoder, reward, continuation, online critic, slow critic, and persistent checkpoint parameters
Study III evaluation	20 sampled calibration episodes per goal/candidate; main 100 sampled + 50 deterministic episodes per goal/condition; 6,750 main episodes
Pairing and integrity	Same goal/mode/reset seeds across conditions; 21/21 calibration and 90/90 main units complete; zero failed attempts; persistent parameter digests unchanged
Stage IV sources	Unmodified 1.25M and 1.50M natural reservoir checkpoints; no actor corruption and no continual retraining
Stage IV conditions	Frozen; standard IR $H = 6$; reward-only IR $H \in \{6, 15, 20\}$; one update per real action; 128 posterior samples; $\alpha = 4 \times 10^{-5}$
Stage IV evaluation	Three goals; 100 sampled + 50 deterministic episodes per cell; 60/60 units and 4,500/4,500 episodes complete on first attempt
Stage IV integrity	Same checkpoint/goal/mode/reset seeds across conditions; no pairing mismatch; persistent parameter digests unchanged; 23.60-hour workflow
Stage V sources	Immutable 1.25M and 1.50M checkpoints; same 384-episode Stage 1A bank; 48,299 factual states and 1,433 valid anchors
Stage V proposals	All five goals; actor-selected and paired recorded actions; 128 Monte Carlo posterior samples; prefix reuse at $H \in \{1, 3, 6, 15, 20\}$
Stage V measurements	Real-state value calibration; online/slow value; full, reward-only, and bootstrap returns; actor-minus-recorded contrasts; continuation and entropy diagnostics
Stage V integrity	124,380 prefix summaries; 18,342,400 actor-imagined transitions per checkpoint; both parameter digests unchanged; canonical specification digest verified

E Complete Study III success table

Table 12: Sampled-policy real-environment success in Study III (100 episodes per cell).

Condition	Kill	Bury	Chop	Mean
Clean actor	.54	.41	.74	.563
Corrupted actor	.36	.06	.43	.283
Standard IR $H = 3$	.52	.04	.51	.357
Standard IR $H = 6$	.44	.03	.41	.293
Standard IR $H = 15$	.32	.01	.30	.210
Standard IR $H = 20$	.55	.02	.46	.343
Entropy-off IR $H = 3$	.48	.03	.32	.277
Entropy-off IR $H = 6$	.49	.03	.22	.247
Entropy-off IR $H = 15$	.46	.05	.35	.287
Entropy-off IR $H = 20$	.34	.05	.42	.270
Reward-only IR $H = 3$	.58	.62	.43	.543
Reward-only IR $H = 6$	.57	.67	.52	.587
Reward-only IR $H = 15$	.64	.96	.59	.730
Reward-only IR $H = 20$	.65	.91	.48	.680
Clean-policy distillation	.58	.14	.73	.483

Table 13: Deterministic-policy real-environment success in Study III (50 episodes per cell). Corruption was calibrated under sampled control and accidentally improved deterministic kill.

Condition	Kill	Bury	Chop	Mean
Clean actor	.32	.42	.74	.493
Corrupted actor	.68	.18	.38	.413
Standard IR $H = 3$	.32	.02	.20	.180
Standard IR $H = 6$	.30	.10	.16	.187
Standard IR $H = 15$	.28	.00	.22	.167
Standard IR $H = 20$	.40	.14	.22	.253
Entropy-off IR $H = 3$	.36	.00	.14	.167
Entropy-off IR $H = 6$	.36	.00	.14	.167
Entropy-off IR $H = 15$	.18	.04	.24	.153
Entropy-off IR $H = 20$	.52	.00	.08	.200
Reward-only IR $H = 3$	.66	.82	.74	.740
Reward-only IR $H = 6$	.60	.90	.74	.747
Reward-only IR $H = 15$	.66	.96	.66	.760
Reward-only IR $H = 20$	.74	.98	.84	.853
Clean-policy distillation	.50	.28	.62	.467

F Complete Study IV success tables

Table 14: Sampled-policy real-environment success in Study IV (100 episodes per cell).

Checkpoint	Condition	Kill	Bury	Chop	Mean
1.25M	Frozen	.56	.71	.84	.703
	Standard IR $H = 6$	.61	.07	.53	.403
	Reward-only IR $H = 6$	.74	.95	.75	.813
	Reward-only IR $H = 15$	.76	.99	.76	.837
	Reward-only IR $H = 20$	.73	1.00	.88	.870
1.50M	Frozen	.23	.14	.03	.133
	Standard IR $H = 6$	.46	.02	.09	.190
	Reward-only IR $H = 6$	.37	.17	.11	.217
	Reward-only IR $H = 15$	.20	.91	.08	.397
	Reward-only IR $H = 20$	.44	.78	.08	.433

Table 15: Deterministic-policy real-environment success in Study IV (50 episodes per cell).

Checkpoint	Condition	Kill	Bury	Chop	Mean
1.25M	Frozen	.64	.36	.78	.593
	Standard IR $H = 6$	.60	.12	.34	.353
	Reward-only IR $H = 6$	.80	.90	.70	.800
	Reward-only IR $H = 15$	.60	.96	.78	.780
	Reward-only IR $H = 20$	.76	.98	.80	.847
1.50M	Frozen	.26	.04	.04	.113
	Standard IR $H = 6$	.36	.00	.12	.160
	Reward-only IR $H = 6$	.24	.12	.14	.167
	Reward-only IR $H = 15$	.38	.86	.04	.427
	Reward-only IR $H = 20$	.18	.78	.10	.353

G Complete Study V diagnostic tables

Table 16: Goal-level real-state online-critic calibration in Study V. The same factual state and target bank is used for both checkpoints.

Checkpoint	Goal	Target	Prediction	Bias	Pearson	Spearman
1.25M	Kill goblin	.198	1.618	+1.421	.182	.121
1.25M	Bury bones	.104	1.725	+1.621	.107	.121
1.25M	Chop logs	.215	.794	+.579	.307	.396
1.50M	Kill goblin	.198	1.991	+1.793	.048	.057
1.50M	Bury bones	.104	2.005	+1.900	.064	.103
1.50M	Chop logs	.215	1.357	+1.142	.061	.082

Table 17: Complete actor-selected factual horizon decomposition in Study V. Means are unweighted across the three trained goals.

Checkpoint	H	Full target	Reward-only	Bootstrap	Bootstrap fraction
1.25M	1	1.443	.052	1.391	.959
	3	1.475	.063	1.412	.952
	6	1.480	.072	1.407	.944
	15	1.482	.097	1.385	.924
	20	1.482	.106	1.376	.917
1.50M	1	1.848	.036	1.812	.975
	3	1.885	.038	1.847	.974
	6	1.891	.039	1.852	.974
	15	1.892	.046	1.846	.971
	20	1.891	.050	1.841	.969

The expanded package contains the original compact Study I tables, Stage II CSV/JSON summaries and vector figures, the complete Study III result, comparison, corruption-selection, and specification files, all 60 Study IV result rows with paired effects, and the complete compact Stage V calibration, target-provenance, actor-contrast, specification, integrity, and figure artifacts. The included `generate_figures.py` regenerates the report-native plots from these audited compact inputs. Raw episode and transition logs remain on the experiment machine and are not duplicated into the report package.

References

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